

PROJECT PPG RECOVERABLE ASSETS (RA) & INVENTORY RISK
MANAGEMENT (IRM)

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A thesis submitted in fulfilment of the
requirements for the award of the degree of

Bachelor of Computer Science (Data Engineering)

Faculty of Computing
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MARCH 2026

DECLARATION

I declare that this thesis entitled “*Recoverable Assets (RA) & Inventory Risk Management (IRM)*” is the result of my own research except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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DEDICATION

This project is dedicated to our families, lecturer, and peers whose continuous support, encouragement, and guidance motivated us throughout the completion of this project. Their contributions have played an important role in helping us achieve the objectives of this work.

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The successful completion of this project would not have been possible without the support and guidance of several individuals and organizations. We would like to express our sincere gratitude to our lecturer for providing valuable guidance, constructive feedback, and continuous support throughout the development of this project.

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ABSTRACT

Inventory risk management is an important process for manufacturing companies because poor inventory visibility can lead to excess stock, obsolete materials, expired inventory, stockout risks, and unnecessary financial losses. This project focuses on developing an end-to-end data engineering pipeline and dashboard for PPG Coatings (Malaysia) Sdn. Bhd. to support Recoverable Assets (RA) identification and Inventory Risk Management (IRM). The main problem addressed in this project is the use of fragmented inventory-related data sources, which makes it difficult for users to obtain a complete and reliable view of inventory status. The aim of this project is to integrate multiple operational datasets, improve data quality, apply inventory risk business rules, and present useful insights through an interactive dashboard. The project adopts the CRISP-DM methodology to guide the overall development process, covering business understanding, data understanding, data preparation, modelling, evaluation, and deployment. Technically, the system is designed using the Medallion Architecture, where raw CSV datasets are ingested into the Bronze layer, cleaned and standardized in the Silver layer, and transformed into analytics-ready datasets in the Gold layer. Microsoft Azure services, including Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, and Azure Synapse Analytics, are used to support data ingestion, storage, transformation, and querying. Microsoft Power BI is then used to develop dashboards that visualize key inventory risk indicators. The system applies business rules to identify materials with stock exceeding twelve months of consumption as Recoverable Assets, detect Magna Carta dead stock with no movement for nine months or more, identify expired inventory, and highlight possible stockout risks based on reorder levels and sales requirements. The outcome of this project is a structured and scalable data pipeline that converts raw inventory data into meaningful business insights. The dashboard helps inventory managers and supply chain managers monitor inventory risks more clearly, reduce manual analysis, and support faster decision-making. Overall, this project contributes a practical data engineering solution that improves inventory transparency, strengthens risk detection, and supports proactive inventory management for PPG Coatings.

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LIST OF ABBREVIATIONS

RA	-	Recoverable Assets
MC	-	Magna Carta
XGBoost	-	Extreme Gradient Boosting
LSTM	-	Long Short-Term Memory
RMDN	-	Recurrent Mixture Density Network
MLP	-	Multilayer Perceptron
1D-CNN	-	One-Dimensional Convolutional Neural Network
SARIMA	-	Seasonal AutoRegressive Integrated Moving Average
AHP	-	Analytic Hierarchy Process
SKU	-	Stock Keeping Unit
AUC	-	Area Under the Curve
SDGs	-	Sustainable Development Goals (by United Nations)
SMOTE	-	Synthetic Minority Over-sampling Technique
ETL	-	Extract, Transform, Load
CRISP-DM	-	Cross-Industry Standard Process for Data Mining
KPIs	-	Key Performance Indicators

CHAPTER 1

INTRODUCTION

1.1 Introduction

This project is built for PPG Coatings (Malaysia) Sdn. Bhd. paints, coatings and surface solutions. The project is to create an end-to-end data pipeline for inventory obsolescence detection and risk management.

The system aims to identify Recoverable Assets (RA), detect obsolete or expired inventory under Magna Carta (MC) classification, and monitor inventory levels to prevent potential stockouts. By integrating data engineering processes with business logic, the project will transform raw operational data into meaningful insights for better inventory decision-making.

1.2 Problem Background

In the current data-centric business workforce, most organizations create extensive data collections which originate from their operational activities that encompass their inventory management, procurement, and productivity. Effective utilization of this data is important to help organizations to run their operations better while maintaining their expenses and making better decisions. However, some organizations out there still face problems in managing and analyzing their inventory data due to fragmented data sources and lack of integrated systems.

Inventory management requires data storage across different datasets which often include materials, inventory snapshots, purchase orders, production records, and sales orders. These datasets generally come with poor integration between them which makes it difficult to focus on obtaining a full view of inventory information. The organizations might also experience difficulties in detecting their main problems such as surplus stock, unused materials, expired materials, shortages and more.

Excess inventory that is also known as Recoverable Assets (RA) refers to products which businesses hold as their backup stock as they exceed their expected usage needs which results in additional storage expenses (Lightspeed., 2025). The identification of dead stock and expired materials at Magna Carta (MC) materials also leads to major financial

losses which arise from these unidentifiable assets. The production system suffers the operational problems due to stockout risk which remains undetectable, resulting in customer dissatisfaction through unfulfilled sales orders (Liu et al., 2025).

The efficient management of inventory data requires organizations to establish a structured method for their data integration process, data processing activities, and data analytics operations. The establishment of an end-to-end data engineering pipeline becomes necessary as it enables organizations to transform their basic data into valuable business insights which improve their inventory risk assessment processes and decision-making activities.

1.3 Project Aim

The project aims to develop an end-to-end data engineering pipeline and dashboard for inventory data analysis and identification of RA, MC materials, and stockout risks.

1.4 Project Objectives

The objectives of this project are:

1. To analyze the inventory risk management problem domain and evaluate current data engineering approaches for detecting recoverable assets, dead stock and stockout risks based on synthetic operational datasets.
2. To design a three-layer data pipeline architecture (Bronze, Silver and Gold layers) that integrates multiple data sources and applies business logic for RA and MC classification.
3. To implement the proposed end-to-end data engineering pipeline using MS Azure, including data ingestion, transformation, storage and processing to generate analytics-ready datasets.
4. To develop and evaluate an interactive dashboard using MS Power BI that visualizes inventory risks and supports decision making by answering key business questions.

1.5 Project Scope

This project focuses on developing an end-to-end data pipeline for inventory obsolescence detection for PPG Industries. The system extracts data from multiple CSV sources, including inventory, materials, purchase orders, production, suppliers, and sales data, followed by data cleaning and transformation using Extract, Transform, Load (ETL)

processes. RA, MC Dead Stock, expired inventory, and stockout risks are to be identified in this project by creating a data pipeline using Microsoft Azure. The processed data is then structured into a data warehouse model for analysis. A dashboard will then be developed using Microsoft Power BI to visualize the inventory status.

1.6 Project Importance

For PPG Coatings to overcome the operational and financial difficulties brought on by fragmented inventory data, this project is essential. The business can reduce needless storage expenses, stop significant financial losses from expired materials, and prevent customer dissatisfaction from unfulfilled sales orders by precisely identifying Recoverable Assets (RA), Magna Carta (MC) dead stock, and stockout risks.

To further accomplish the project's objective of converting unprocessed, compartmentalized operational data into useful business insights, an end-to-end data engineering pipeline is essential. The system enables stakeholders to shift from reactive tracking to proactive decision-making by centralizing, processing, and visualizing this data using Microsoft Azure and Power BI. This ultimately optimizes overall inventory risk management.

1.7 Report Organization

This report is organized into distinct chapters to provide a structured presentation of the project.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter explores the academic and technical literature concerning the creation of data engineering systems, with a specific focus on managing inventory and detecting risks. The primary objective of this discussion is to build a thorough understanding of the theoretical bases, methodological strategies, and technological structures that support the architecture and execution of the suggested system.

The review starts by investigating the principles of supply chain management and the significance of enterprise resource systems within contemporary inventory workflows. It continues by evaluating the Medallion Architecture as an organized method for designing data pipelines, followed by an inquiry into the concepts and instruments of data visualization that pertain to inventory analytics. Additionally, a comparative study of previous research is included to pinpoint current deficiencies in the sector which the proposed solution intends to resolve. The chapter also analyzes Microsoft Azure as the selected cloud environment and Microsoft Power BI as the central tool for visualization in this initiative.

Through a critical assessment of these subjects, the development of the comprehensive data engineering pipeline can be firmly established upon recognized expertise and validated techniques. The knowledge acquired throughout this chapter offers the essential guidance needed to create a successful remedy for the inventory obsolescence and risk management difficulties faced by PPG Coatings Malaysia Sdn. Bhd.

2.2 Supply Chain Management

Supply chain is a system comprising people, processes and technologies that deliver value to customers (Stanton, 2020). This value can be in terms of products, services, financial flows or information (Mentzer et al., 2001). The ultimate goal of supply chain management is to increase profitability and minimise cost. It coordinates procurement of raw materials, inventory management, production and distribution.

There are three types of expenses associated with inventory systems, including ordering

costs; inventory holding costs such as storage charge, insurance and interest; and shortage cost that involves loss of sales and customer goodwill. The inventory management problem is maintaining sufficient stock to meet demand while balancing the holding costs and stockout risk (Singh & Kumar, 2011).

Inventory of the supply chain goes through multiple stages, including procurement, storage, usage and finally obsolescence. There are several risks in inventory management. This includes excess inventory, which ties up capital; obsolete or expired inventory, which leads to financial write-offs; and stockouts that affect operations and customer orders. The common strategies to mitigate the risks includes Just-in-Time (JIT) inventory (Balkhi et al., 2022), Economic Order Quantity (EOQ) (Senthilnathan, 2019) and Safety stock planning (Kırmızı et al., 2024) .

Obsolete inventory is stock that a business can no longer sell or use due to lack of demand, product lifecycle completion or market changes. Dead stocks are stocks that have no movement over a period of time. Excess stock is inventory exceeding expected demand. These stocks gradually transform from an asset into a liability which requires write down or provisions. Holding inventory is not free, it is related to the capital and incurs storage cost. Therefore, most companies need provisions for the loss in value of non-performing assets.

Stockout is the lack of the required item in the inventory. Shortage of raw materials can disrupt production, while the lack of products will cause dissatisfaction for customers and lost sales. Several factors influence stockout risk, including lead time variability, safety stock levels and reorder point policies

2.3 Enterprise Resource System

The data engineering pipeline in an Enterprise Resource Planning (ERP) system defines a systematic method which gathers, merges, modifies, and distributes the data from various organizational departments to create a single data analysis platform. The system generates multiple types of data which necessitate data pipelines to maintain a uniform and dependable data transmission between various system components. The pipeline starts when the data is collected from many sources such as transactional databases, IoT devices, and external applications. The data then undergoes a transformation process

which includes cleaning, normalization, and aggregation before it is stored in data warehouses or data lakes.

The modern ERP pipeline uses ETL and ELT frameworks to create standardized data sets which they will use for analytics purposes. Metadata-driven methods help to increase scalability and maintainability of data as they allow multiple enterprise systems to be integrated in a more flexible manner (Vattumilli, 2024). The combination of big data technologies with cloud computing systems allows the ERP pipelines to handle both batch and real-time data processing which results in faster and more precise decision-making (Agarwal, 2024). The pipeline often uses machine learning techniques to develop predictive analytics models and systems that can detect unusual patterns.

However, there are several challenges which include problems with data quality, difficulties in system integration, and limitations in data scalability. The organizations need to build strong pipeline systems which include data governance, monitoring, and automation features as these systems will help the organizations to provide dependable performance and assist them for decision-making processes (Zakaria et al., 2024).

2.4 Medallion Architecture

To ensure data quality and consistency, modern data pipelines often adopt a multi-layered approach known as the Medallion Architecture. This architectural framework organizes data into progressive layers, allowing raw data to be systematically refined into high-quality, analytics-ready datasets. By separating data processing into distinct stages, the Medallion Architecture enhances data governance, scalability and reliability while supporting efficient downstream analysis. Figure 2.4.1 shows the Medallion Data Architecture.



Figure 2.4.1 Medallion Data Architecture (DataForge, 2018)

Medallion Architecture contains three layers :

Bronze Layer (Raw Data): It serves as the destination for unprocessed, raw data that is taken straight from the source systems such as CSV files with inventory snapshots. Here, speedy ingestion and historical archiving without changing the initial data state are the main priorities.

Silver Layer (Cleansed and Conformed Data): The raw data is cleaned, filtered, and standardized in this layer. A consolidated view of operations is produced by handling missing values and joining fragmented tables.

Gold Layer (Business-Level Data): The last layer is extremely sophisticated and analytics-optimized. The classification rules for Recoverable Assets (RA), Magna Carta (MC) dead stock, and stockout risk calculations are examples of specific business logic that is applied here. The final dataset will be dependable, organized, and ready for business intelligence tools to use right away thanks to this layered approach.

Through this approach, the Medallion Architecture ensures that data progresses from raw data to refined in a controlled and structured manner, producing reliable, high-quality data that can be successfully used for analytics and strategic insights.

2.5 Data Visualization

Data visualization involves transforming data into graphical formats like charts, graphs, and dashboards. Within inventory risk management, this practice acts as a vital link connecting intricate analytics with practical business strategies. It allows business leaders to spot trends, irregularities, and patterns within massive datasets swiftly and naturally. Good visualization adheres to core concepts like interactivity, accuracy, clarity, and consistency.

This guarantees that team members can keep track of key performance indicators, such as expiration dates, stock amounts, and reorder limits, even if they lack extensive technical skills (Charan et al., 2024).

Various visual formats are frequently utilized within supply chain and inventory operations. Time series line graphs monitor inventory volumes over different periods to uncover usage patterns and seasonal changes. Meanwhile, heatmaps and bar charts contrast stock amounts across different product units to pinpoint areas of high risk involving outdated or slow moving goods. By linking these visuals to a cloud based data pipeline, the graphics update automatically to show the latest inventory status. This creates the continuous monitoring required for preemptive risk control.

For this initiative, Microsoft Power BI acts as the main visualization platform because it integrates flawlessly with Microsoft Azure services and offers extensive interactive features. Power BI natively accommodates numerous visual styles and permits the creation of advanced calculated metrics using the DAX formula language, which helps analysts construct dashboards without heavy programming. The platform can easily combine data from various origins, such as Azure Synapse Analytics and Azure Data Lake. This capability makes it an excellent fit for the Medallion Architecture pipeline utilized in this work, eventually allowing the team at PPG Coatings Malaysia to transition from passive inventory observation to anticipatory decision making (Charan et al., 2024).

2.6 Microsoft Azure

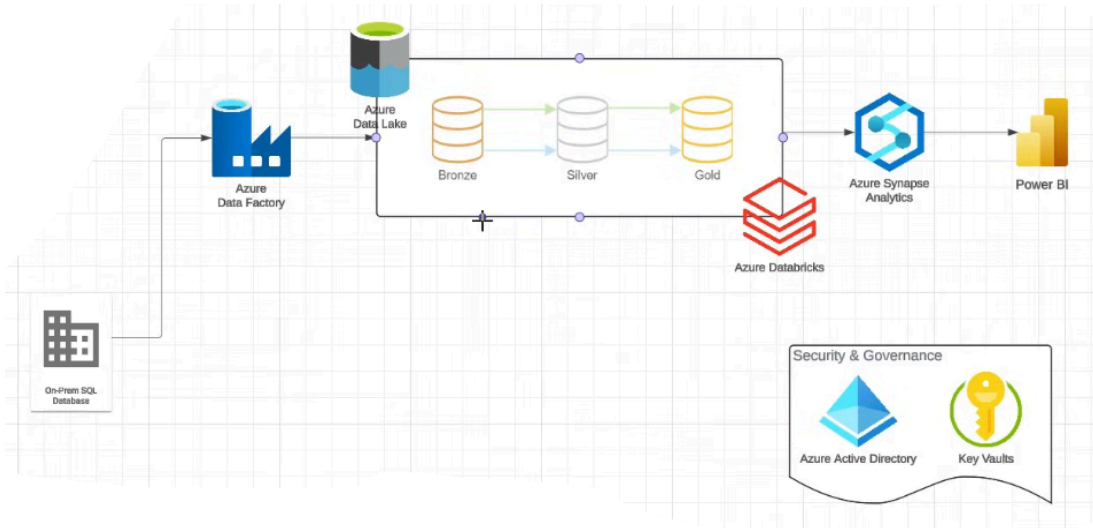


Figure 2.6.1 CLASSIC Microsoft Azure data architecture diagram (Microsoft Learn, 2016)

This is a graphical presentation of a Modern Data Analytics Architecture on Microsoft Azure based on the given diagram (Riscutia, 2021). It shows the overall data flow between an on-premise source system and a final visualization dashboard with a data processing architecture based on a "Medallion Architecture" (Bronze, Silver, and Gold Layer) (Armbrust et al., 2021). Here is a step-by-step breakdown of the components and the data flow:

The On-Prem SQL Database is the starting point of the pipeline as it is a traditional, local database, in which the raw business data is generated. Then, Azure Data Factory, a data integration service in the cloud, is the orchestrator (Swinbank, 2021). It retrieves the raw data on the on-premises database and transfers it to the cloud storage environment of an ETL or ELT process.

After extraction, the data first rests in the Azure Data Lake, a centralized and scalable cloud data storage repository (Nader, 2022). The data is then subjected to a rational process of data refinement in three data layers. The Bronze Layer contains the unverified data in the same form that it was consumed in the source. This is then converted into the Silver Layer where the data is cleaned, filtered and converted into a standardized format. Lastly, it is transferred to the Gold Layer, which consists of the most refined and aggregated information that is presented in business reporting and analytics format. All of this transformation is facilitated by the Apache Spark-based analytics solution, Azure

Databricks, which will provide the compute power to read the information in the Data Lake and push it through the Bronze, Silver and Gold layers with ease (Chambers & Zaharia, 2018).

Once the data is refined to the standard of Gold, it is loaded into Azure Synapses Analytics (Rubiolo et al., 2020). This service will serve as the enterprise data warehouse and will be a high-performance querying engine on the completed data. In order to transform this information into action, Power BI, a business intelligence tool, is directly integrated with Azure Synapse analytics (Powell, 2018). This enables end-users to develop and engage with detailed dashboards, reports, and visual insights on the data processed.

Basic security and management is upheld throughout the pipeline with dedicated Azure services. Azure Active Directory is an identity and access controller, so only authorized users and services can access the data architecture (Nickel, 2019). In conjunction with this, Key Vaults are used to store sensitive data, like passwords, database connection strings, and encryption keys, in a secure environment, so they do not have to be hard coded into the pipeline configuration (Diogenes & Shinder, 2018).

2.7 Microsoft Power BI

Microsoft Power BI is a business intelligence and data visualization tool that is used to transform raw data into interactive dashboards and reports. It supports data integration from multiple sources such as databases and cloud platforms, allowing users to analyse large datasets efficiently (Charan et al., 2024). Power BI provides features such as data modelling, real-time analysis and customizable dashboards, which aids in monitoring KPIs and identifying trends. In this project, Power BI is used to visualise inventory data and support the detection of obsolescence, including RA and MC materials. Compared to traditional reporting tools, it offers better flexibility, ease of use and integration with Azure, making it suitable for data-driven decision making systems.

2.8 Chapter Summary

This chapter created the theoretical and technological foundations which enable the creation of a data pipeline that will identify inventory obsolescence at PPG industries. The research studied fundamental principles of data pipelines together with inventory lifecycles and automated ETL processes which organizations require to maintain operational control. The literature review demonstrated how systems evolved from basic rule-based systems to AI-driven models which identify the particular gaps this project will solve. The proposed technology stack which includes Microsoft Azure, Medallion Architecture, and Power BI shows how these integrated tools create a strong foundation which enables PPG to move from reactive monitoring to proactive inventory risk mitigation.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodology used for the development of the inventory obsolescence detection and risk management system. A structured and data-driven approach is essential to ensure that both the analytical objectives and system implementation are aligned with business needs. Therefore, the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework is selected as the methodology. The six phases of CRISP-DM which are business understanding, data understanding, data preparation, modeling, evaluation, and deployment provides a guideline to transform raw data into meaningful insights while allowing continuous improvement throughout the project.

In terms of the technical implementation, the Medallion Architecture is used to build the end-to-end data pipeline. The Medallion Architecture is used to manage and transform large volumes of fragmented inventory data into structured, analytics-ready datasets. Furthermore, tools such as Azure Data Factory, Azure Databricks, Azure Synapse Analytics and Microsoft Power BI are used for data processing and visualization. This chapter explains how these technologies are mapped to the CRISP-DM phases.

3.2 Methodology

For this project, the CRISP-DM was selected for this project because it is suitable for data-driven system development. It consists of structured phases such as business understanding, data understanding, data preparation, modeling, evaluation and deployment, which align well with the development of the inventory obsolescence detection system. This methodology helps with tasks including detecting a RA and identifying MC obsolete materials by emphasis on extracting useful insights from data. Instead of sticking to a set linear sequence, CRISP-DM allows for interactive improvements and adjustments based on data findings, making it more suitable than Waterfall. Overall, it provides both a systematic and flexible approach, making it easier to refine the data pipeline and improve accuracy throughout the project (Tripathi et al., 2021; Nava & Hernández, 2012).

3.3 Phases of the Methodology

CRISP-DM comprises six iterative phases from business understanding to deployment.

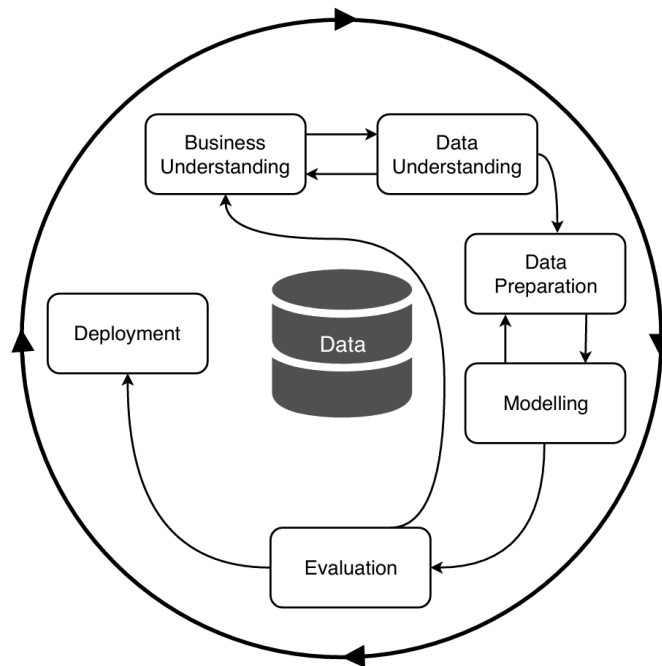


Figure 3.1 The CRISP-DM process model of data mining (Martinez-Plumed et al., 2019)

The first phase involves **business understanding**. In developing any solution, it is critical to understand the problem context and the needs of the business. This can be achieved by drawing a clear data mining goal that serves as a baseline for the future processes. It also involves assessing available resources and required resources, as well as setting a structured project plan (Schröder et al., 2021). In this project, stakeholder engagement is conducted to obtain the business requirements.

After understanding the business, the next phase is **data understanding**. This phase involves data collection and data exploration. Data quality is assessed and descriptive analysis is conducted to gain initial insights (Schröder et al., 2021). If necessary, the data mining goals may be revised and refined to better suit the available data.

The next phase is **data preparation**. This stage prepares the final dataset for modelling. It involves data selection, cleaning, integration, transformation and feature engineering (Schröder et al., 2021). In this project, the Medallion Architecture approach is adopted, where the data is refined through Bronze, Silver and Gold layers.

Subsequently, **modelling** is conducted. Various modelling techniques are experimented to determine the most suitable algorithms to address the business problem (Schröder et al., 2021). These algorithms are evaluated using domain knowledge, success criteria and test

design. Depending on the data needs, it may be necessary to revisit the data preparation to obtain more suitable attributes and features.

Evaluation is conducted before deployment to ensure that the results align with the business objectives. The results and the entire pipeline are evaluated (Schröer et al., 2021) to determine whether to proceed with deployment or to iterate further.

Finally, the deployment phase ensures that the model delivers actionable results to the stakeholders. The developed model or pipeline can result in a final report or a software component. It involves planning the deployment, monitoring and maintenance (Schröer et al., 2021). In this project, a dashboard will be produced to present the findings.

3.4 Technology used description

Although the general project lifecycle is based on the **CRISP-DM approach** (Schroer et al., 2021), technical implementation of the data pipeline is based on the standard data engineering phases using a **Medallion Architecture** (Armbrust et al., 2021). In this section, we will describe the particular technologies, platforms, and tools that will be used to deploy the end-to-end data pipeline and inventory risk management system. The project is mainly developed based on the Microsoft Azure cloud platform, using its scalable data services and **Microsoft Power BI** as an analytics visualization tool.

In order to provide a clear picture of how the project management methodology can be applied to the technical implementation of the system, the **CRISP-DM** phases are directly mapped to the standard data engineering phases and the Microsoft Azure tools. This alignment will guarantee that each technical step will be used to reach a particular project milestone.

3.4.1 Technology Used Description

Table 3.4.1.1 shows the description of technology used in this project.

Table 3.4.1.1 Technology Used Description

CRISP-DM Phase	Data Engineering Phase	Azure Implementation	Tools/Technologies Used
----------------	------------------------	----------------------	-------------------------

Data Understanding & Data Preparation	Ingestion & Storage (Bronze & Silver Layers)	Raw data is ingested into the Bronze layer, then cleaned and transformed into the Silver layer.	Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks
Modeling	Processing (Gold Layer)	Business rules and analytics are applied and stored in the Gold layer for optimized querying.	Azure Databricks, Azure Synapse Analytics
Deployment & Evaluation	Serving, Visualization & Security	Data is visualized via dashboards and secured with identity and key management systems.	Microsoft Power BI, Microsoft Entra ID, Azure Key Vault

3.4.2 Detailed Technology Description

Table 3.4.1.2 shows the details of technology description in this project.

Table 3.4.1.1 Detailed Technology Description

No.	Technology	Description
1	Azure Data Factory	Primary data integration and orchestration tool for ETL workflows.
2	Azure Data Lake Storage Gen2	Centralized storage with Bronze, Silver, and Gold layers.
3	Azure Databricks	Core processing engine using Apache Spark for transformation and analytics.
4	Azure Synapse Analytics	High-performance SQL data warehouse for analytics queries.
5	Microsoft Power BI	Interactive dashboards and data visualization platform.
6	Azure Key Vault	Security and governance through identity and secret management.

3.5 Project Planning Schedule

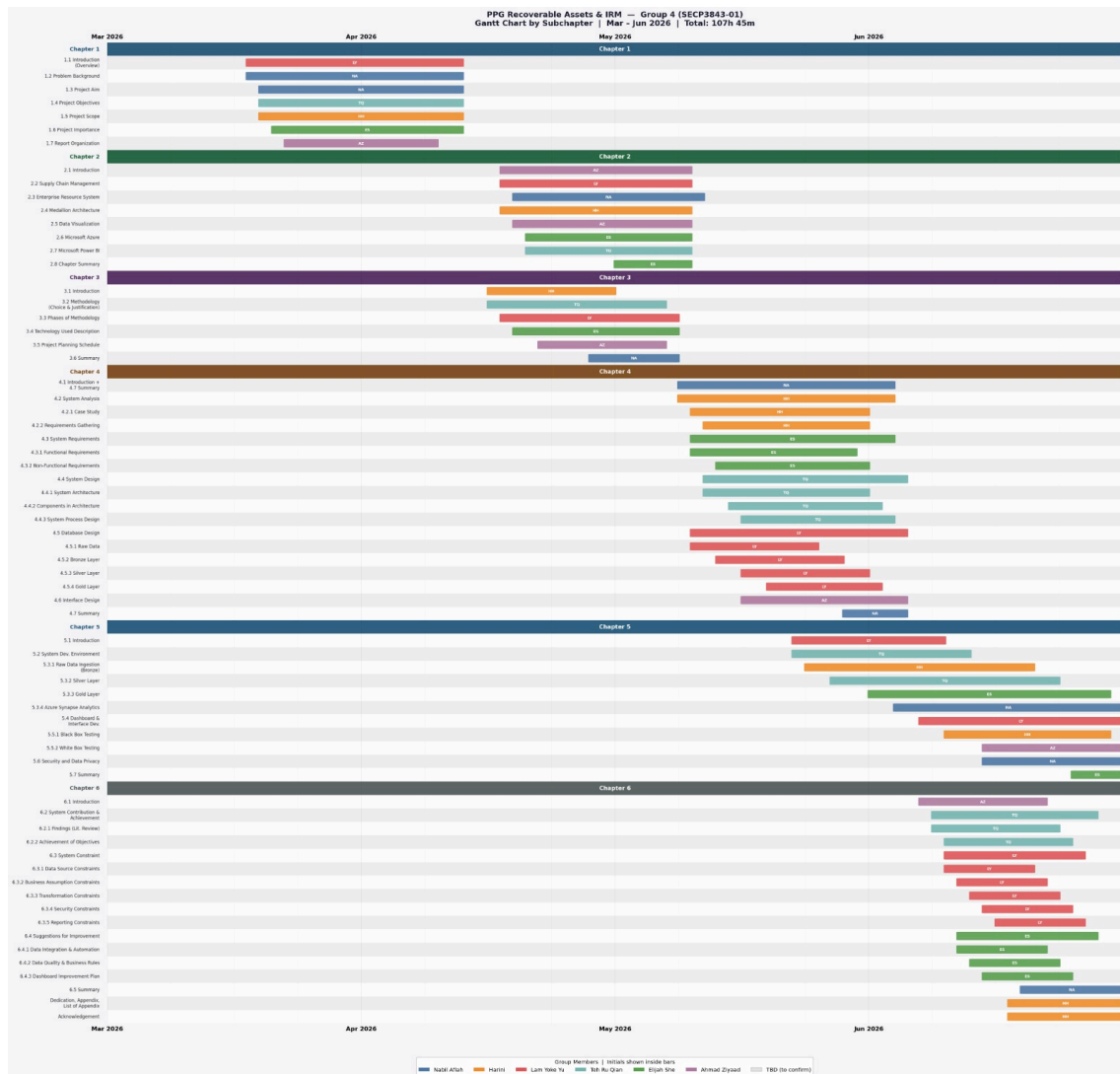


Figure 3.5.1 Gantt Chart

Figure 3.5.1 illustrates the project planning schedule for the PPG Recoverable Assets and Inventory Risk Management project as a Gantt chart. The timeline spans from March 2026 to June 2026, with a total accumulated effort of 107 hours and 45 minutes across all six group members. The project is structured into six chapters, with each subchapter assigned to individual members to ensure clear accountability and an even distribution of workload.

Chapter 1 on Introduction was completed in March 2026, covering subchapters such as the Problem Background, Project Aim, Objectives, Scope, Importance, and Report Organization. Chapter 2 on Literature Review followed from mid-March to end of April 2026, with members contributing to topics including Supply Chain Management, Enterprise Resource Systems, Medallion Architecture, Data Visualization, Microsoft

Azure, Microsoft Power BI, and the Chapter Summary.

Chapter 3 on Methodology was completed in April to early May 2026, encompassing the Methodology Choice and Justification, Phases of Methodology, Technology Used Description, and the Project Planning Schedule. Chapter 4 on Analysis and Design was the most extensive phase, running from late April through May 2026. It covered System Analysis, Requirements Gathering, Functional and Non-Functional Requirements, System Architecture, System Process Design, Database Design across the Raw, Bronze, Silver, and Gold layers, Interface Design, and the Chapter Summary.

Chapter 5 on Implementation and Testing ran from mid-May to early June 2026, covering the data pipeline implementation across all three Medallion layers, Azure Synapse Analytics, Dashboard Development, Black Box and White Box Testing, and Security and Data Privacy. Chapter 6 on Conclusion was completed from late May through mid-June 2026, addressing System Contributions, Achievement of Objectives, System Constraints, Suggestions for Improvement, and supporting sections such as the Appendix and Acknowledgements. Each member is represented by a distinct colour in the Gantt chart, with initials shown inside the bars. Unassigned tasks are marked as TBD.

Overall, the schedule reflects a logical and progressive project structure. Foundational chapters were completed first, followed by more technical and evaluative work in the later stages. This sequencing is consistent with the CRISP-DM methodology adopted by the team, where business understanding and design decisions were established before implementation began.

3.6 Summary

This chapter demonstrates designation and implementation of the proposed data engineering solution by adopting CRISP-DM framework, which consists of six crucial phases that includes business understanding, data understanding, data preparation, modeling, evaluation, and deployment phase. This chapter also provides a detailed description of building a complete data pipeline through the Medallion Architecture which helps to organize the data into three layers that are bronze, silver, and gold to enhance both data quality and processing capabilities. The implementation of the end-to-end data engineering solution relies on cloud-based tools which include Azure Data Factory for data ingestion and orchestration, Azure Databricks for data

transformation, Azure Synapse Analytics for data storage and analytical processing, and Microsoft Power BI for interactive dashboards development process to visualize data insights and support for PPG industry decision-making.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 Introduction

This chapter discusses the system analysis and design to analyse the business requirements and design the overall structure of the proposed system to ensure the solution that will be developed is capable of supporting the inventory monitoring, risk detection, and analytical reporting in an effective way. The data was collected during industrial visit to PPG to learn more about their inventory process, business rules, and the more detailed analytical products that are expected. Information collected was used to determine both functional and non-functional requirements of the system so that the proposed solution meets the project objectives and user needs. In addition, this chapter also explains the architectural and structural design of the proposed system, which includes the system architecture, system components, schema of the database, the process workflows, and the user interface design. The Medallion Architecture is also detailed indicating the organization and flow of data over the raw data, bronze, silver, and gold layers. Several modeling diagrams have been included in this chapter to give a complete vision of the interactions between the system and users involved. Overall, this chapter provides a solid basis for the analysis and design stage of the system to guarantee all technical specifications are taken care of before the implementation phase.

4.2 System Analysis

This section presents the analysis of the current business environment and system needs for the proposed solution. It focuses on understanding existing inventory management, identifying key issues, and determining how a data-driven system can address these challenges. The analysis also explains how requirements were gathered and refined.

4.2.1 Case Study

This part focuses on understanding the current inventory management at PPG Coatings and identifying gaps that can be improved through a data-driven solution. The existing inventory process relies on fragmented data sources and manual monitoring. Currently, inventory-related data is stored across multiple CSV files, including materials, inventory snapshots, purchase orders, production orders, suppliers, and sales orders.

Although these datasets contain valuable operational information, they are not integrated into a centralized system. As a result, users such as inventory managers and supply chain managers face difficulties in obtaining a unified view of inventory status. This leads to challenges in identifying slow-moving or obsolete stock and tracking material consumption trends. In addition, there is limited analytical capability in the current system. The insights are not presented in an interactive or user-friendly format. Therefore, decision-making is based on static data or manual calculations, which reduces efficiency and accuracy. The absence of a structured data pipeline also affects data quality, as inconsistencies such as missing values, duplicate records, and non-standardized formats are not properly managed.

As a solution, the proposed system introduces a structured data pipeline based on the Medallion Architecture, integrating data ingestion, transformation, and analytics into a unified platform. The system aims to automate data processing, improve data quality, and enable real-time risk detection. By implementing business rules such as recoverable asset identification, dead stock detection, and stockout risk analysis, the system provides actionable insights to support operational and strategic decision-making.

4.2.2 System Requirements Gathering Techniques

The initial project scope and problem statement were provided prior to the industrial visit, and the visit was utilised to validate assumptions and obtain further clarification on certain requirements.

Before the industrial visit, preliminary understanding was developed based on the problem description and the datasets provided by the stakeholder. This includes identifying key objectives such as inventory risk detection, data integration, and dashboard reporting. Early analysis of the CSV files was also conducted to understand the data structure, attributes, and potential relationships between datasets.

During the industrial visit to PPG Coatings, clarification discussions were carried out to confirm system expectations. For example, specific criteria for identifying Magna Carta Dead Stock, expired inventory, and stockout risks were verified during the visit. These discussions also helped reporting needs and key performance indicators required for decision-making.

4.3 System Requirements

System requirements are the description of the functions and quality characteristics that the proposed PPG Inventory Risk Management System must fulfill. The requirements are split into functional and non-functional requirements. Functional requirements specify what the system needs to do. Non-functional requirements define the qualities the system should have, e.g. reliability, security, scalability and usability.

4.3.1 Functional Requirements

The functional requirements of the proposed system are shown in Table 4.3.1.1.

Table 4.3.1.1 Functional Requirements

Requirement ID	Requirement	Description
FR1	Data Ingestion	The system shall ingest six CSV datasets, including materials, inventory snapshot, purchase orders, production orders, suppliers, and sales orders.
FR2	Raw Data Storage	The system shall store the original raw datasets in the Bronze layer without modifying the source data.
FR3	Data Cleaning and Transformation	The system shall clean and transform the raw data by removing duplicates, handling missing values, standardizing date formats, and validating data types.
FR4	Recoverable Asset Identification	The system shall identify materials with stock exceeding 12 months of consumption as Recoverable Assets.
FR5	Magna Carta Dead Stock Detection	The system shall detect Magna Carta materials with no movement for 9 months or more.

FR6	Expired Inventory Detection	The system shall identify inventory items that have exceeded their expiry dates.
FR7	Stockout Risk Detection	The system shall detect materials with insufficient stock based on reorder levels and open sales order requirements.
FR8	Financial Risk Calculation	The system shall calculate the financial impact of obsolete, expired, and dead stock inventory.
FR9	Dashboard Generation	The system shall generate interactive Power BI dashboards to display inventory risk indicators and support decision-making.
FR10	User Role Support	The system shall support different user roles, including Data Engineer, Inventory Manager, and Supply Chain Manager.

4.3.2 Non Functional Requirements

The non-functional requirements of the proposed system are shown in Table 4.3.1.2.

Table 4.3.1.2 Non Functional Requirements

Requirement ID	Requirement	Description
NFR1	Reliability	The system should process data consistently and accurately without data loss during ingestion, transformation, and loading.
NFR2	Scalability	The system should be able to support increasing data volume and additional inventory-related datasets in the future.
NFR3	Data Quality	The system should ensure that the processed data is clean, complete,

		standardized, and suitable for reporting and analysis.
NFR4	Security	The system should protect sensitive information by using secure authentication, role-based access control, and secret management.
NFR5	Usability	The dashboard should present inventory risks clearly so that non-technical users can understand and interpret the results easily.
NFR6	Maintainability	The system should be modular and easy to update when there are changes in business rules, data sources, or reporting requirements.
NFR7	Performance	The system should process data and produce analytics-ready outputs within an acceptable time for business reporting.
NFR8	Auditability	The system should preserve raw data in the Bronze layer to support future checking, reprocessing, and validation.

Overall, these requirements define the main system functions and quality expectations needed to support inventory risk detection, data processing, and dashboard-based decision-making for PPG Coatings Malaysia.

4.4 System Design

This section explains the overall system design, focusing on how the data flows through the system, the technologies involved, the main system modules, the inputs and outputs and the overall ETL process.

4.4.1 Data Architecture

The proposed system architecture starts with six CSV datasets provided by PPG. These datasets serve as the **input layer** of the system and contain operational data related to materials, inventory, suppliers, purchasing, production and sales.

The first stage of the architecture is the **data ingestion layer**, where Azure Data Factory is used to extract data from the CSV files and move the data to Azure Data Lake Storage Gen2. The raw data is stored in the Bronze layer, where the original files remain unchanged for auditing and backup purposes.

The second stage is the **data processing layer**, where Azure Data Factory is used to clean, validate and transform the raw data. This stage forms the Silver layer, where missing values are handled, duplicate records are removed, data formats are standardized. While business logic is applied in the Gold layer. The business logic includes:

- Identifying materials with stock exceeding 12 months consumption (Recoverable Assets)
- Detecting Magna Carta Dead Stock (no movement for 9+ months)
- Identifying expired inventory
- Calculating total financial write-down provisions
- Detecting stockout risks
- Identifying affected customer sales orders

The third stage is the **data warehouse layer**, where transformed data is loaded into Azure Synapse Analytics under the Gold layer. This layer stores structured, analytics-ready data using a star schema model to improve reporting performance.

The final stage is the **visualization layer**, where Microsoft Power BI connects to Azure Synapse Analytics to generate dashboards and reports. Users such as inventory managers and supply chain managers can monitor inventory risks and make business decisions based on real-time insights.

For security purposes, Microsoft Entra ID is used for user authentication and role-based access control.

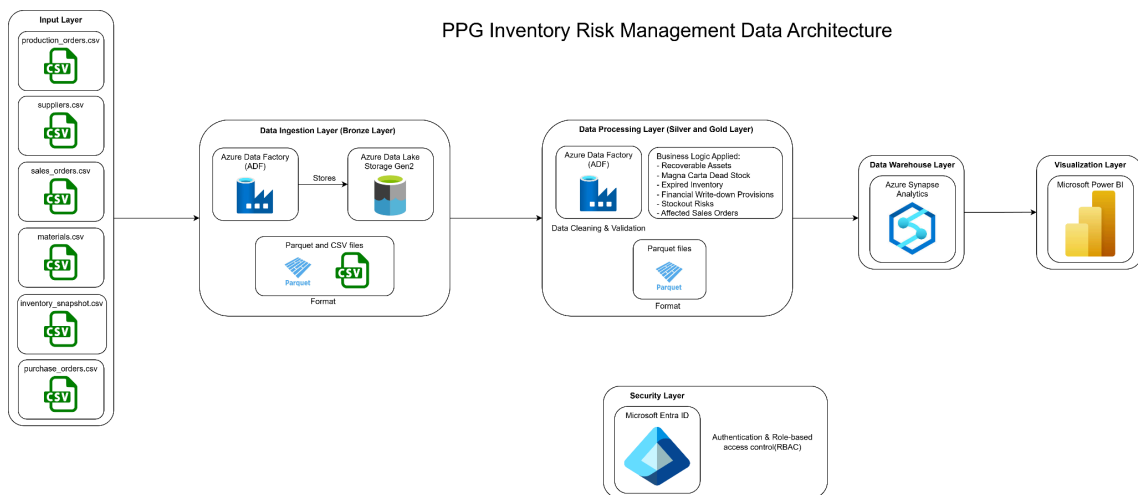


Figure 4.4.1.1 Data Architecture

4.4.2 Components in data architecture

Data Source Layer

This layer consists of 6 CSV files which provide information about inventory, supplier, production and sales needed for analysis.

Azure Data Factory (Bronze, Silver and Gold Layer)

Azure Data Factory extracts and loads data into the raw data storage. This layer also performs data transformation and business rule implementation and is stored into Azure Synapse Analytics. It helps to schedule data pipelines and ensures data is transferred efficiently.

Azure Data Lake Storage Gen2 (Bronze Layer)

This layer stores raw data exactly as received. It acts as a backup layer and allows future reprocessing when required.

Azure Synapse Analytics

This layer stores processed data in structured tables optimized for business intelligence reporting.

Power BI

Power BI visualizes processed data through dashboards and reports.

Microsoft Entra ID

Used to authenticate users and manage access permissions.

4.4.3 System Process Design

The proposed system follows the **ETL** methodology. In the **Extract Phase**, Azure Data

Factory extracts data from the 6 source files and transfers them into Azure Data Lake Storage. During the **Transform Phase**, Azure Databricks processes raw data by removing duplicates, handling missing values, validating data types, applying inventory risk logic and generating calculated fields. Finally, in the **Load Phase**, processed data is loaded into Azure Synapse Analytics and connected to Power BI dashboards.

Use Case Diagram

The use case diagram shows the interaction between users and PPG Inventory Risk System. This system involved 3 actors:

- Data Engineer: Responsible for uploading raw datasets and executing the ETL pipeline.
- Inventory Manager: Responsible for monitoring inventory risks and detecting dead stock items.
- Supply Chain Manager: Responsible for analyzing the impact of sales orders and generating dashboard reports for operational decision making.

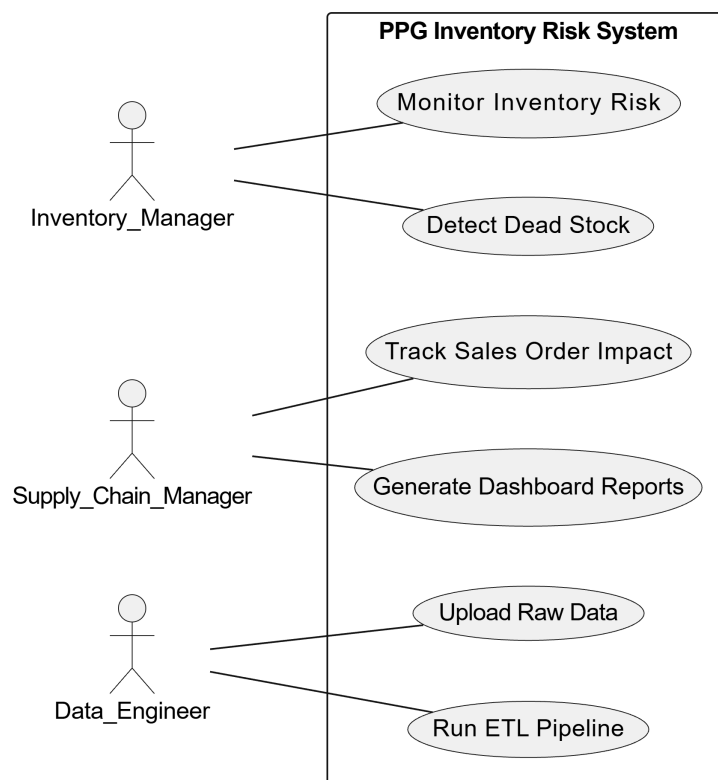


Figure 4.4.3.1 Use Case Diagram

Sequence Diagram

The sequence diagram shows the activities performed by the Data Engineer during ETL

and analytics workflow.

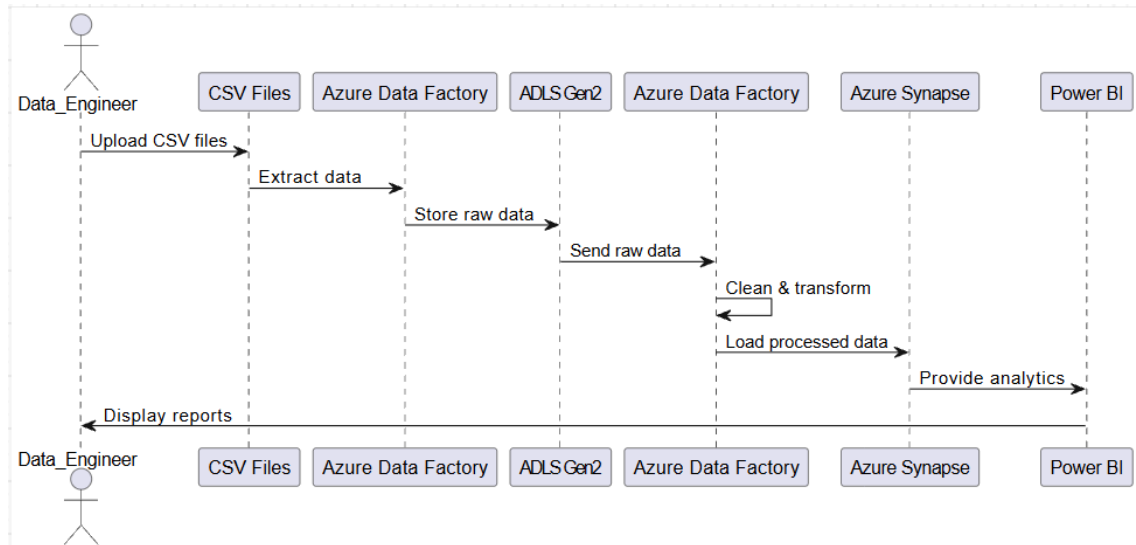


Figure 4.4.3.2 Sequence Diagram

Activity Diagram

This diagram shows the workflow of the system.

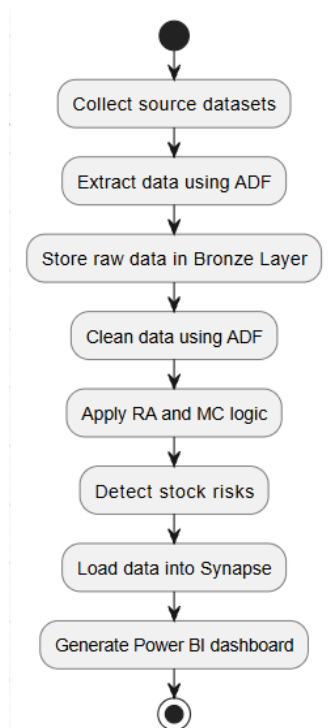


Figure 4.4.3.3 Activity Diagram

4.5 Database Design

The database design follows the medallion architecture with bronze, silver and gold containers. The raw data will be ingested into the bronze container without any modification. At this layer, all the raw data will be reserved in its original form. Then, the data will be cleaned and transformed, stored in the silver container. Simple modelling will be conducted to ensure data integrity and relationship between data. Subsequently, the data will be curated and analytical data will be derived, forming the gold layer data. This layer includes details that are analytical ready. The following diagram shows the overall structure of the database architecture.

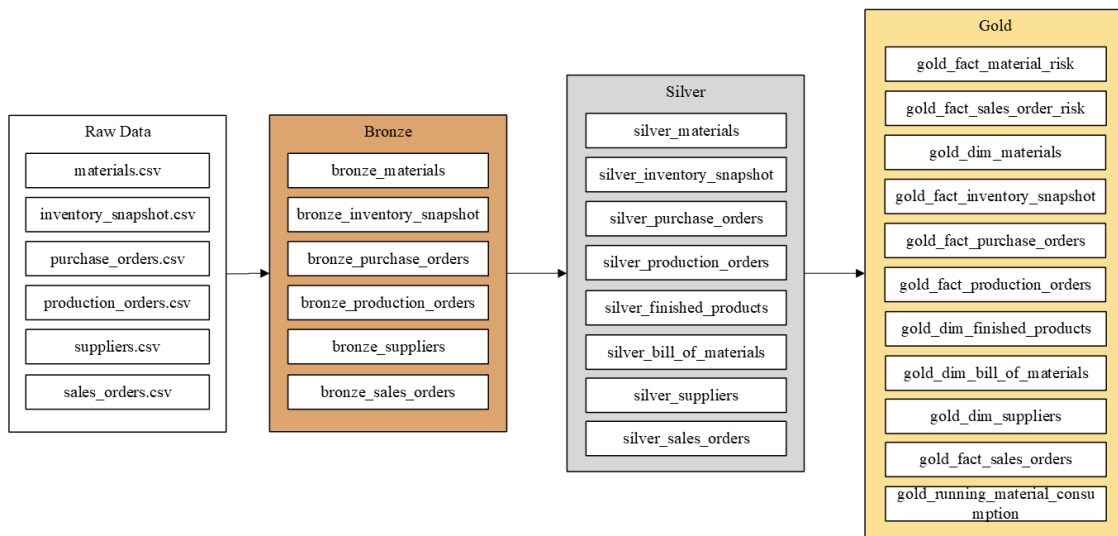


Figure 4.5.1: Overview of the database under the Medallion Architecture

4.5.1 Raw Data

The following table is the raw data files that will be used for this project.

Table 4.5.1: Raw data provided by stakeholder

Files	Description
materials.csv	20 raw materials across 4 categories: Pigment, Resin, Solvent, Packaging
inventory_snapshot.csv	Monthly stock levels per material, including lot expiry dates
purchase_orders.csv	Purchase orders from 10 suppliers, including late and partial deliveries
production_orders.csv	Material consumption records for 10 finished paint products
suppliers.csv	Supplier reference data: name, country, reliability rating
sales_orders.csv	Customer sales orders, including open orders at risk due to shortage

4.5.2 Bronze Layer

In this layer, the raw data will be ingested into the data lake. Each of the csv files will be stored as bronze_*.csv in the Bronze container created in the storage. Basic data quality checks without any modification will be conducted, including null detection and type validation.

Table 4.5.2: Bronze Layer Data Dictionary

Table	Attributes	Data Type	Example Data
bronze_inventory_snapshots	snapshot_date	Date	31/01/2025, 28/02/2025
	material_id	String	MAT001, MAT002
	warehouse_location	String	WH-A, WH-B
	quantity_on_hand	Integer	380, 600
	lot_expiry_date	Date	31/01/2025, 28/02/2025
bronze_materials	material_id	String	MAT001, MAT002
	material_name	String	Titanium Dioxide, Iron Oxide Red
	category	String	Pigment, Resin
	unit_of_measure	String	KG, Litre, Unit
	reorder_level	Integer	200, 1000
	lead_time_days	Integer	7, 10
	unit_cost	Float	8.5, 3.45
bronze_production_orders	production_order_id	String	PRD-001, PRD-002
	production_date	Date	15/01/2025, 07/02/2025
	finished_product	String	Anti-Rust Coating, Automotive Red Paint
	material_id	String	MAT001, MAT002
	quantity_consumed	Integer	240, 300
bronze_purchase_order	po_id	String	PO-001, PO-002
	po_date	Date	23/01/2025, 01/11/2025
	supplier_id	String	SUP006, SUP002
	material_id	String	MAT001, MAT002
	quantity_ordered	Integer	2000, 1500
	quantity_received	Integer	2000, 0
	expected_delivery_date	Date	23/01/2025, 01/11/2025
	actual_delivery_date	Date	23/01/2025, 01/11/2025
bronze_sales_orders	sales_order_id	String	SO-001, SO-002

	order_date	Date	23/01/2025, 01/11/2025
	customer_name	String	BuildRight Hardware, Kerjaya Bina Bhd
	finished_product	String	Premium White Paint 4L, Anti-Rust Coating
	quantity_ordered	Integer	618, 107
	required_delivery_date	Date	23/01/2025, 01/11/2025
	status	String	Delivered, Open
bronze_suppliers	supplier_id	String	SUP001, SUP002
	supplier_name	String	ChemBase Sdn Bhd, SinoCoat Materials Co
	country	String	Malaysia, Singapore
	reliability_rating	String	High, Medium, Low

4.5.3 Silver Layer

In this layer the data are cleaned and stored with the same structure as the bronze layer without much modification, where bronze_* tables will be transformed into silver_* tables and stored in the silver container.

Operations include standardising data types and date format and removing duplicates. In addition, a data integrity check will be conducted to ensure all foreign keys are not orphaned. Besides, a silver_finished_products table and a silver_bill_of_materials table are created. These tables are derived from the production_orders.

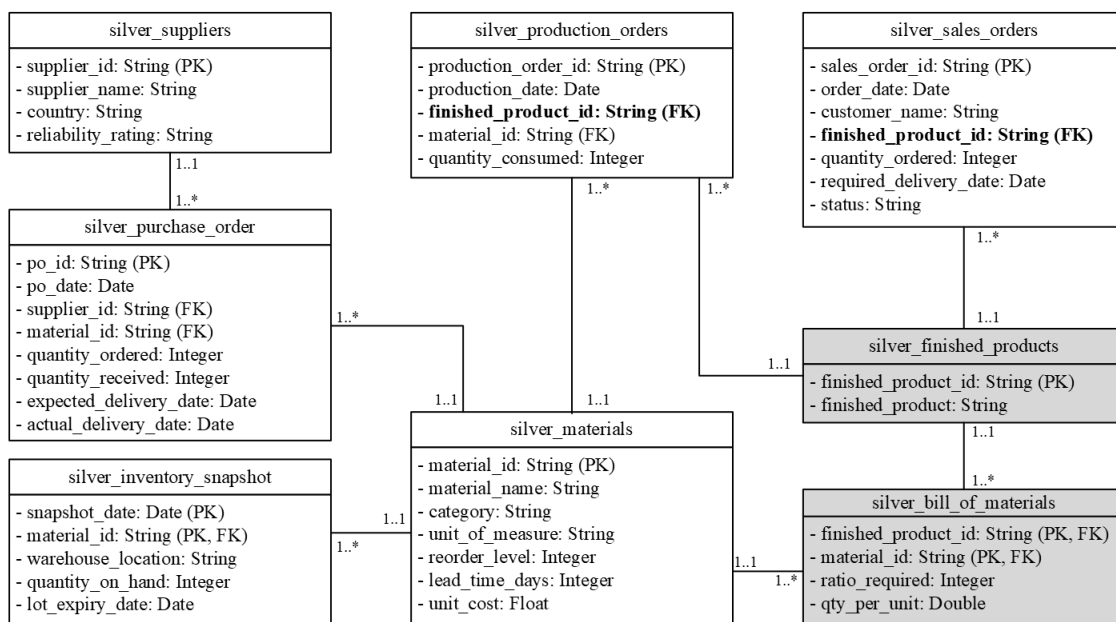


Figure 4.5.2: Entity Relation Diagram for the data under the Silver Layer

Since there is no exact amount of the finished product provided for each production order, in the bill of materials (BOM), the quantity per unit is determined by the total production consumption divided by the total delivered sales quantity. It is assumed that all production orders are made to fulfill an order or to replenish the inventory of the delivered products.

4.5.4 Gold Layer

In this layer, the data are filtered and new attributes that provide insights to the stakeholders will be derived. The entities from the silver layer will be used to form a star constellation schema. The fact tables and dimension tables will be renamed as gold_fact_* for fact tables or gold_dim_* for dimension tables.

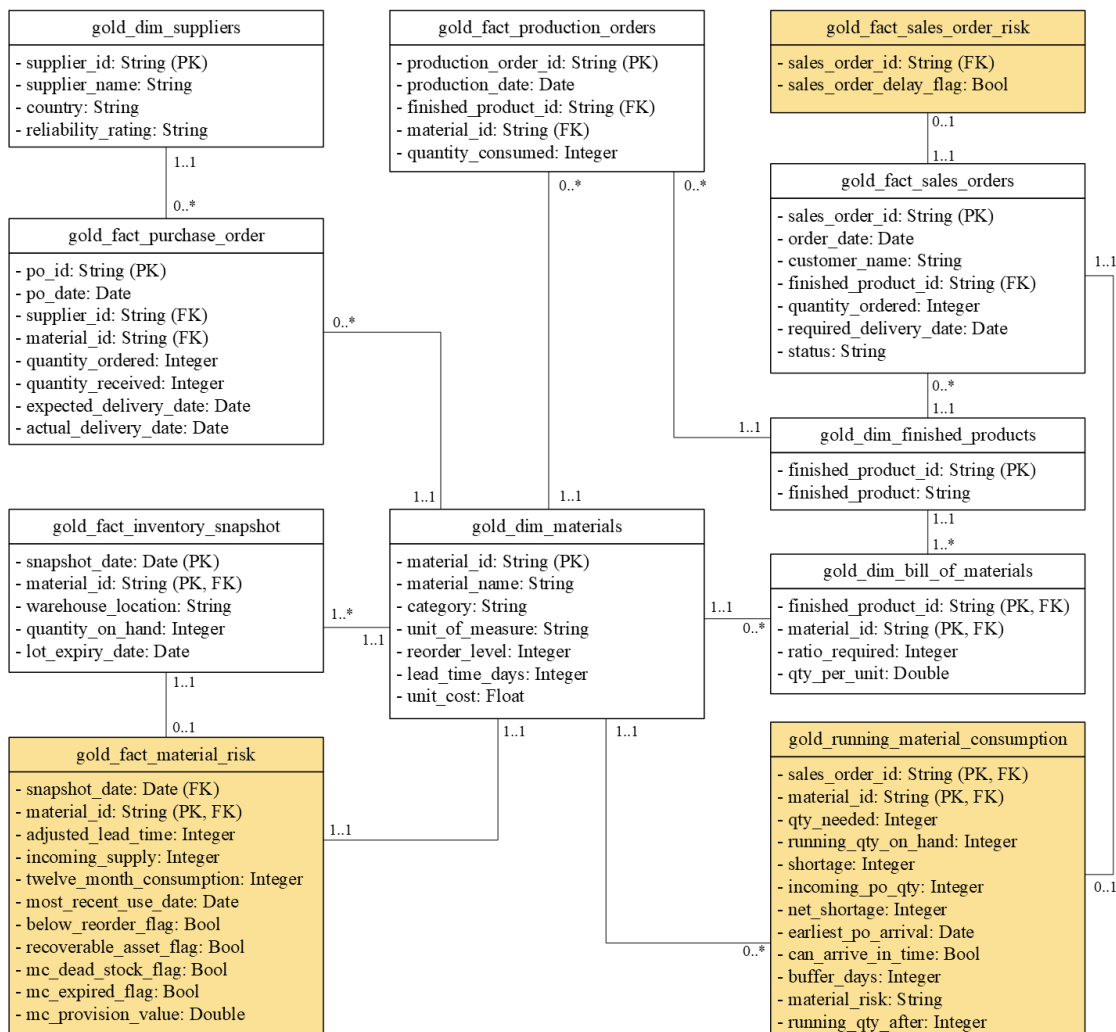


Figure 4.5.2: Entity Relation Diagram for the data under the Gold Layer

From the data, the business insights are determined, forming the material risk data and

sales order risk data. The newly added data aims to facilitate reporting and visualisation.

The `gold_fact_material_risk` data are derived from the `inventory_snapshot`, `materials`, `production_orders` to provide insights about the Magna Carta Dead Stock and provision value, Magna Carta Expired, Recoverable Assets, and materials under reorder level.

Table 4.5.3: Data Dictionary for `gold_fact_material_risk`

Attribute	Description	Derivation
<code>snapshot_date</code>	Latest snapshot date for the material	-
<code>material_id</code>	Unique identifier of the material	-
<code>adjusted_lead_time</code>	Lead time adjusted based on historical delivery delays of the material	<code>materials.lead_time + AVG(purchase_order.actual_delivery_date - purchase_order.expected_delivery_date)</code>
<code>incoming_supply</code>	Total quantity of the material in the purchase order that is not fulfilled yet	<code>SUM (purchase_order.purchase_quantity WHERE purchase_order.actual_delivery_date == null)</code>
<code>twelve_month_consumption</code>	The total quantity of material used over the 12 months of production.	<code>SUM (production_order.quantity_consumed WHERE production_date >= current_date - 12 months)</code>
<code>most_recent_use_date</code>	Latest usage of the material according to the <code>production_orders</code>	-
<code>below_reorder_flag</code>	True if the material is below the reorder level.	<code>inventory_snapshot.qty_on_hand < materials.reorder_level</code>
<code>recoverable_asset_flag</code>	True if the material where stock on hand exceeds expected 12-month consumption	<code>inventory_snapshot.qty_on_hand > twelve_month_consumption</code>
<code>mc_dead_stock_flag</code>	Flagged true for Magna Carta Dead Stock if the material is not consumed for the last 9 months	<code>most_recent_use_date < (current_date - 9 months)</code>
<code>mc_expired_flag</code>	Flagged true for Magna Carta Expired if the lot is expired	<code>inventory_snapshot.lot_expiry_date < current_date</code>
<code>mc_provision_value</code>	The provision value if the material is flagged as Magna Carta Expired	<code>IF mc_dead_stock → inventory_snapshot.qty_on_hand * materials.unit_cost ELSE → 0</code>

A supporting table, `running_material_consumption` table is created to check the availability of materials and if materials shortage can be replenished before the required delivery date of the sales order.

It is assumed that all materials for a sales order is reserved for the sales order and will not be used for the coming sales orders.

Table 4.5.4: Data Dictionary for `gold_running_material_consumption`

Attribute	Description	Derivation
<code>sales_order_id</code>	Unique identifier of the sales_order	-
<code>material_id</code>	Unique identifier of the material	-
<code>qty_needed</code>	Quantity of materials needed for the sales order	<code>sales_order.quantity_ordered * bom.qty_per_sales_unit</code>
<code>running_qty_on_hand</code>	Inventory at time of evaluation	IF first order for this material → <code>inventory_snapshot.quantity_on_hand</code> ELSE → previous row's <code>running_qty_after</code> for this material
<code>shortage</code>	Raw gap	<code>running_qty_on_hand - qty_needed</code>
<code>incoming_po_qty</code>	Incoming supply before deadline	SUM outstanding POs where <code>expected_delivery_date ≤ required_delivery_date</code>
<code>net_shortage</code>	Gap after incoming POs	<code>shortage + incoming_po_qty</code>
<code>earliest_po_arrival</code>	Order arrival estimate if ordered today	<code>today + adjusted_lead_time_days</code>
<code>can_arrive_in_time</code>	If the PO can arrive in time if ordered today.	<code>earliest_po_arrival ≤ required_delivery_date</code>
<code>buffer_days</code>	To determine the days to the delivery or the days of delay	<code>required_delivery_date - earliest_po_arrival</code> (negative for delayed sales_orders)
<code>material_risk</code>	Risk of materials, if its is available, blocked or tight	IF <code>net_shortage ≥ 0</code> → OK ELSE IF <code>can_arrive_in_time = FALSE</code> → BLOCKED ELSE IF <code>buffer_days ≤ 5</code> → TIGHT ELSE → OK
<code>running_qty_after</code>	Stock passed to next order	<code>running_qty_on_hand - qty_needed</code>

According to the material consumption, the sales order risk can be determined. A sales_order_risk table will be used to store the delay flag.

Table 4.5.5: Data Dictionary for gold_sales_order_risk

Attribute	Description	Derivation
sales_order_id	Unique identifier of the sales_order	-
sales_order_delay_flag	Flagged TRUE if the order will be delayed	IF ANY material_risk = BLOCKED for this sales_order_id → TRUE ELSE → FALSE

4.6 Interface Design

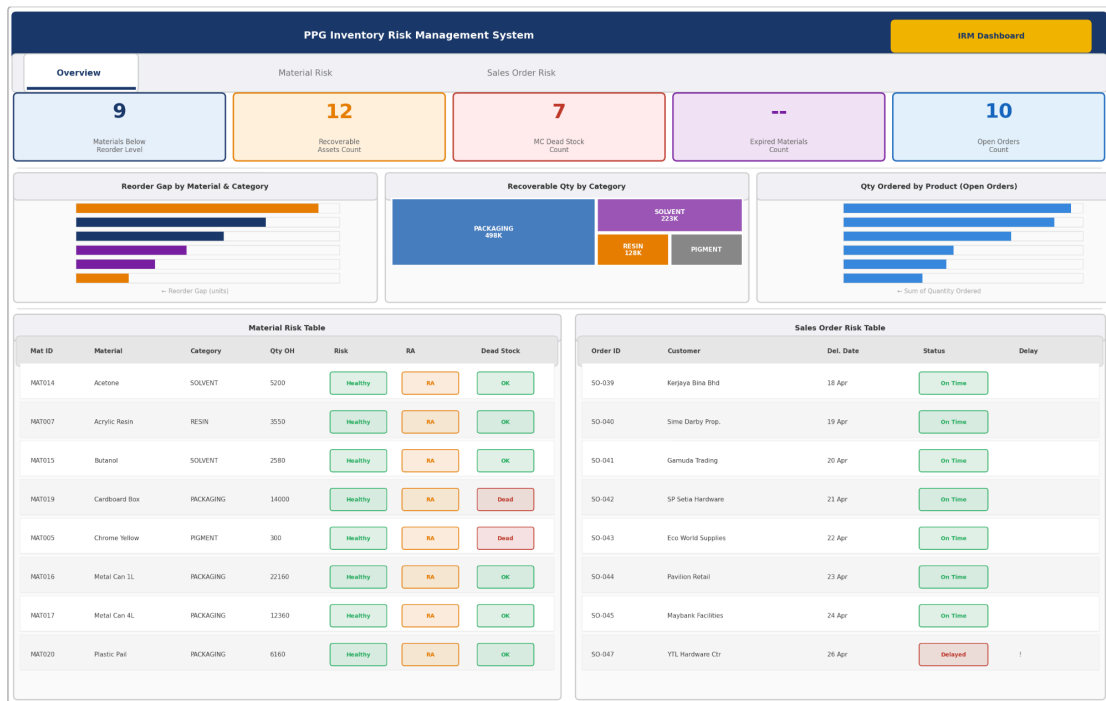


Figure 4.6.1: Interface Design for PowerBI using Python matplotlib

The interface design for the PPG Inventory Risk Management system is presented as a wireframe prototype in Figure 4.6, illustrating the proposed layout of the Microsoft Power BI dashboard before implementation. The dashboard consists of three pages accessible through a tab-based navigation bar, namely Overview, Material Risk, and Sales Order Risk. Figure 4.6 depicts the Overview page, which serves as the primary landing view for all user roles.

At the top of the page, five KPI card visuals are arranged horizontally across the canvas, each colour-coded by risk type. The cards display the count of materials below reorder

level, recoverable assets, MC dead stock items, expired lots, and open sales orders respectively. This colour differentiation allows users to distinguish between risk categories at a glance.

Below the KPI row, three chart visuals are placed side by side. The first is a horizontal bar chart showing the reorder gap by material, the second is a treemap displaying recoverable asset quantities distributed across product categories, and the third is a horizontal bar chart presenting quantity ordered per finished product for open orders.

The lower section contains two detail tables presented side by side. The Material Risk Table lists each material with its risk classification flags rendered as colour-coded badges. The Sales Order Risk Table lists open orders with their delivery status, where delayed orders are highlighted in red to draw immediate attention. Together, these components provide a complete inventory risk overview within a single dashboard page.

4.7 Summary

To summarize, this chapter focuses on the detail of system analysis and design required for creating the inventory obsolescence detection, mapping the business logic to the technical requirements that relate the raw CSV data to business analytical solutions. The significance of this chapter is to make sure that the data pipeline has the architectural capability to be able to accurately measure the financial commitments and stockout challenges that are critical to maintaining inventory health. The guidelines and frameworks that have been set in this chapter are a blueprint for the implementation process. The system defines the flow of data across the bronze, silver, and gold layers so that data stays clean and organized throughout the flow and into a schema that is optimized for Power BI implementation. Finally, these design specifications and diagrams form the basis for the development process in the next chapter, which ensures the solution of this project will give the actionable insights as PPG needs.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Introduction

This chapter presents the implementation and testing of the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The implementation focuses on building the end-to-end data pipeline using Microsoft Azure services, starting from raw data ingestion, data transformation, and storage into analytics-ready datasets.

The chapter also explains how the processed data is used to support dashboard development in Microsoft Power BI for inventory risk monitoring. In addition, testing activities are carried out to ensure that the pipeline, business rules, and dashboard outputs function correctly and meet the project requirements. Overall, this chapter shows how the system design is converted into a working solution that supports inventory risk detection and decision-making.

5.2 System Development Environment

The development of the data pipeline system for PPG Industries was carried out by Microsoft Azure and Power BI for data visualization. Table 5.2.1 summarises all tools and technologies used throughout the system development.

Table 5.2.1 System Development Environment

Tool/Technology	Justification
Azure Data Lake Storage Gen2 (ADLS Gen2)	Central data storage for all pipeline layers:bronze, silver and gold containers stored in storage account 'ppgdatalake01'
Azure Data Factory (ADF)	Primary pipeline orchestration and transformation tool used to build and execute all bronze, silver and gold layer dataflow activities
Azure Synapse Analytics	Analytical query engine used to create 'gold_db' database, map gold layer Parquet files to SQL views, execute business analytics queries and design the star schema
Power BI Desktop	Dashboard and report development tool used to design and build interactive visualisations connected to 'gold_db' database
Apache Parquet	Columnar storage format used for all bronze, silver and gold layer output files stored in ADLS Gen2
CSV (Comma-Separated Values)	Source file format for all six raw input datasets ingested into the pipeline

SQL (T-SQL)	Query language used within Azure Synapse Analytics Serverless SQL Pool for creating views, executing analytics queries and validating star schema relationships
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5.3 Data Pipeline Implementation

The data pipeline was implemented using Microsoft Azure services to enable efficient data ingestion, processing, and visualization. Azure Data Factory was used to extract data from source systems and load it into Azure Data Lake Storage. The data lake follows a layered architecture (Bronze, Silver, Gold), where raw data is stored, transformed, and refined into analytics-ready datasets. Azure Databricks performs data processing and transformation, while Azure Synapse Analytics supports querying and data warehousing. Finally, Microsoft Power BI is used to create interactive dashboards for data visualization.

5.3.1 Raw Data Ingestion (Bronze Layer)

Azure Data Factory was used to perform the raw data ingestion. Data was extracted from multiple sources of CSV files and loaded into the data lake, where they are stored in the Bronze layer. Storing the raw data into the Bronze layer ensures the data is preserved for traceability and future processing.

The pipeline ingests six datasets, each representing different aspects of the inventory and supply chain operations:

- **materials.csv** contains information on 20 raw materials categorized into Pigment, Resin, Solvent, and Packaging.
- **inventory_snapshot.csv** stores monthly stock levels for each material, including lot expiry dates, which are important for identifying expired inventory.
- **purchase_orders.csv** includes procurement records from 10 suppliers, containing order quantities, delivery timelines, as well as late and partial deliveries.
- **production_orders.csv** records material consumption data for the production of 10 finished paint products.
- **suppliers.csv** provides supplier reference details such as supplier name, country, and reliability rating.
- **sales_orders.csv** contains customer sales order data, including open orders that may be affected by material shortages.

Once the datasets were ingested, minimal processing was performed to maintain the

integrity of the raw data while ensuring basic data reliability. The processing steps focused on data quality checks, including null value detection, data type validation, and duplicate handling. Figure 5.3.1.1 shows the data quality checks performed on the production_orders dataset.

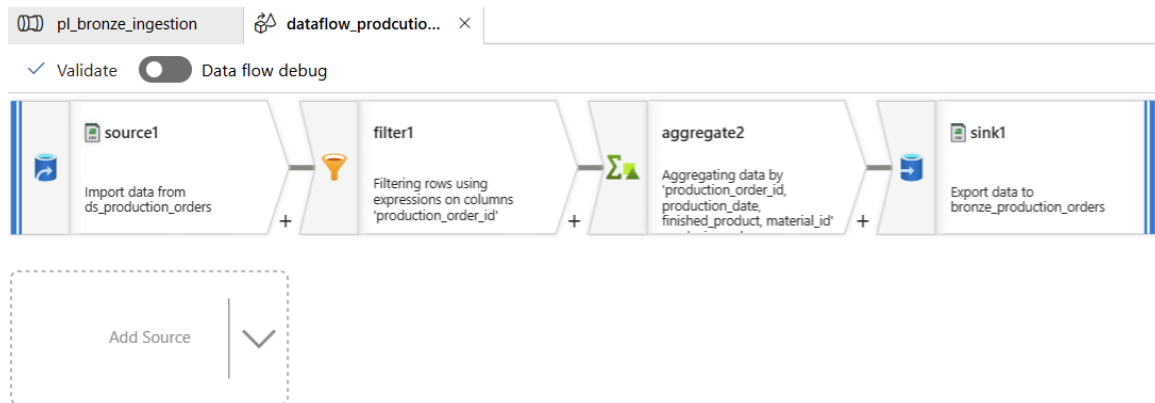


Figure 5.3.1.1 : Data Quality Checks Performed On production_orders Dataset

After these checks, the minimally processed data was stored in the Bronze layer of the data lake. Each dataset was saved in both CSV and Parquet formats, where CSV files preserve the original structure of the raw data, while Parquet files provide optimized storage and improved performance for downstream processing.

5.3.2 Silver Layer

In the Silver layer, data from the Bronze layer is cleaned, standardized and validated before being stored in a structured format. This layer is to ensure data consistency and quality across all tables, making the data suitable for analytical processing in the Gold layer. The tasks carried out in this layer include standardising data types and date formats, trimming and normalizing string values, removing duplicate records, enforcing foreign key integrity checks to remove orphaned records and deriving new tables from existing data. Azure Data Factory was used for all transformations and the processed data was stored in the silver container of Azure Data Lake Storage Gen2 in Parquet format.

The Silver layer pipeline, named pl_silver, was set up to execute eight Data Flow activities in structured order. As shown in Figure 5.3.2.1, the first three dataflows: 'df_silver_materials', 'df_silver_suppliers' and 'df_silver_finished_products' are executed in parallel as they have no interdependencies. The next dataflows, namely 'df_silver_inventory_snapshot', 'df_silver_purchase_order', 'df_silver_production_orders' and 'df_silver_sales_orders' are executed after the first

three complete successfully as they depend on the reference tables produced in the earlier stage. Finally ‘df_silver_bill_of_materials’ is executed last as it depends on both ‘silver_production_orders’ and ‘silver_sales_orders’.

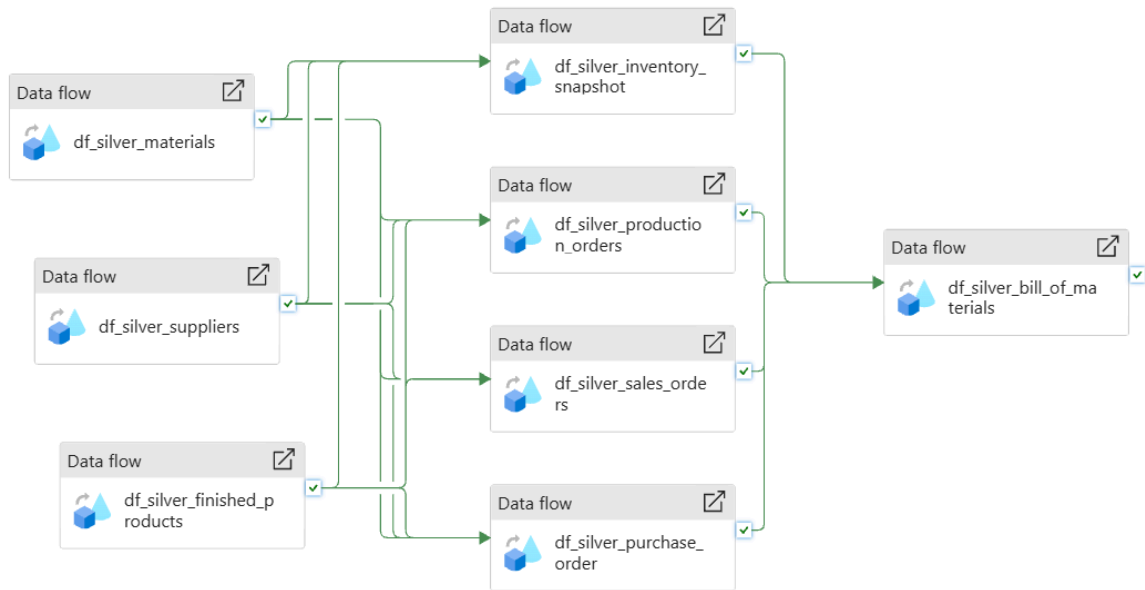


Figure 5.3.2.1 Silver layer pipeline

df_silver_materials

As shown in Figure 5.2.3.2, the ‘df_silver_materials’ dataflow reads from the ‘bronze_materials’ dataset. In the Derived Column transformation, string columns such as ‘material_id’ and ‘material_name’ were trimmed of leading and trailing whitespace using the ‘trim()’ function. While ‘category’ and ‘unit_of_measure’ were additionally converted to uppercase using the ‘upper()’ function. The ‘unit_cost’ was cast to float and rounded to two decimal places using ‘round(toFloat(unit_cost), 2)’. Then, a Filter transformation was applied to remove any rows where the primary key column ‘material_id’ was null. To remove duplicate records, an Aggregate transformation was configured to group by ‘material_id’ and apply the ‘first()’ function on all remaining columns, retaining the first occurrence of each unique material. The cleaned and deduplicated data was then written to the ‘silver_materials’ dataset in the silver container.

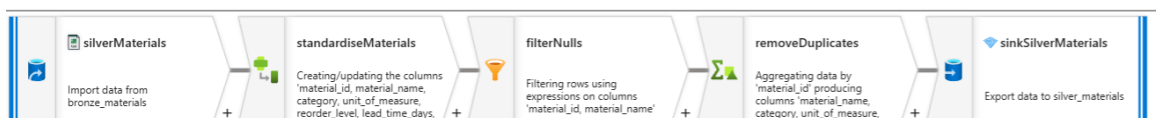


Figure 5.2.3.2 df_silver_materials

df_silver_suppliers

Figure 5.2.3.3 shows the ‘df_silver_suppliers’ dataflow reads from the ‘bronze_suppliers’ dataset. In the Derived Column transformation, ‘supplier_id’ and ‘supplier_name’ were

trimmed using the 'trim()' function. While 'country' was converted to uppercase using 'upper(trim())' to standardise country names across all records. The 'reliability_rating' column was trimmed to remove any surrounding whitespace. A Filter transformation was applied to exclude rows where 'supplier_id' or 'supplier_name' was null. An aggregate transformation grouped records by 'supplier_id' and applied 'first()' on all other columns to eliminate any duplicate supplier entries. The processed data was written to the 'silver_suppliers' dataset.

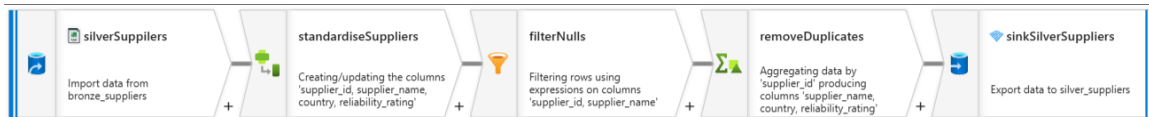


Figure 5.2.3.3 df_silver_suppliers

df_silver_finished_products

In Figure 5.2.3.4, the 'df_silver_finished_products' dataflow is a derived table created from the 'bronze_production_orders' dataset. Since the source data does not contain a dedicated finished product identifier column, a surrogate key was generated using the 'crc32()' hash function applied to the finished product name, prefixed with "FP-" to produce a readable and consistent identifier. The expression used was 'concat('FP-', toString(crc32(trim(toString(finished_product)))))'. This approach ensures that the same product name always produces the same identifier, maintaining referential integrity across all tables that reference finished products. A Select transformation was applied to retain only the 'finished_product_id' and 'finished_product' columns, discarding all other production order columns. A Filter transformation removed rows with null values in either column and an Aggregate transformation grouped by 'finished_product_id' with 'first(finished_product)' to produce a unique dimension table of finished products.



Figure 5.2.3.5 df_silver_finished_products

df_silver_inventory_snapshot

Figure 5.2.3.6 shows the 'df_silver_inventory_snapshot' dataflow reads from the 'bronze_inventory_snapshot' dataset, which captures periodic stock levels across warehouse locations. The Derived Column transformation standardised the 'snapshot_date' column by converting it from string to date format using 'toDate(toString(snapshot_date), 'yyyy-MM-dd')', cast 'quantity_on_hand' to integer and converted 'warehouse_location' to uppercase using 'upper(trim())'. A Filter

transformation removed rows where either 'snapshot_date' or 'material_id' was null, as these form the composite primary key of this table. An Aggregate transformation grouped by both 'snapshot_date' and 'material_id' to remove duplicate snapshot entries, applying 'first()' on all other columns. An Exists transformation was then applied to enforce foreign key integrity, retaining only rows where 'material_id' exists in the 'silver_materials' table. This was configured with 'silver_materials' as the right stream and the condition 'material_id' == 'material_id.' The validated data was written to the 'silver_inventory_snapshot' dataset.

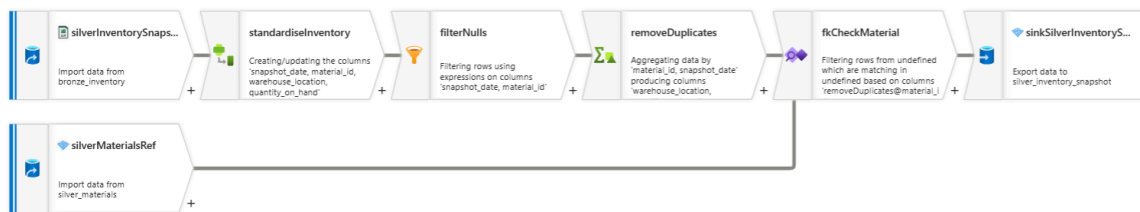


Figure 5.2.3.6 df_silver_inventory_snapshot

df_silver_purchase_order

Figure 5.2.3.7 shows the 'df_silver_purchase_order' dataflow reads from the 'bronze_purchase_orders' dataset. The Derived Column transformation standardized all string identifiers using 'trim()', converted date columns 'po_date', 'expected_delivery_date' and 'actual_delivery_date' from string to date format using 'toDate()' and cast 'quantity_ordered' and 'quantity_received' to integer. A Filter transformation retained only rows with non-null values in 'po_id', 'supplier_id' and 'material_id'. An Aggregate transformation grouped by 'po_id' and applied 'first()' on all other columns to deduplicate records. Two Exists transformations were then applied sequentially. The first checked that 'supplier_id' exists in 'silver_suppliers'. The second checked that 'material_id' exists in 'silver_materials' to ensure full referential integrity across both foreign key relationships. The processed data was written to the 'silver_purchase_order' dataset.



Figure 5.2.3.7 df_silver_purchase_order

df_silver_production_orders

Figure 5.2.3.8 shows the 'df_silver_production_orders' dataflow reads from the 'bronze_production_orders' dataset and applies similar standardisation as other dataflows. In addition to trimming string columns and converting 'production_date' to date format and 'quantity_consumed' to integer, a new 'finished_product_id' column was derived using the same 'crc32' expression applied in 'df_silver_finished_products', ensuring consistent identifier generation across both tables. A Filter transformation removed rows with null values in 'production_order_id', 'finished_product' or 'material_id'. An Aggregate transformation grouped by 'production_order_id' and applied 'first()' on all other columns. Two Exists transformations were applied. One to verify 'finished_product_id' against 'silver_finished_products' and another to verify 'material_id' against 'silver_materials'. A Select transformation was applied to retain only the five columns defined in the ERD schema: 'production_order_id', 'production_date', 'finished_product_id', 'material_id' and 'quantity_consumed', discarding the original 'finished_product' name column. The validated data was written to the 'silver_production_orders' dataset.



Figure 5.2.3.8 df_silver_production_orders

df_silver_sales_orders

Figure 5.2.3.9 shows the 'df_silver_sales_orders' dataflow reads from the 'bronze_sales_orders' dataset. The Derived Column transformation trimmed all string identifiers, converted 'order_date' and 'required_delivery_date' from string to date format, cast 'quantity_ordered' to integer, converted 'status' to lowercase using 'lower(trim())' to standardise status values and generated 'finished_product_id' using the same 'crc32' expression for consistency. A Filter transformation removed rows where 'sales_order_id', 'finished_product' or 'order_date' was null. An Aggregate transformation grouped by 'sales_order_id' and applied 'first()' on all other columns. An Exists transformation verified that each 'finished_product_id' has a matching record in 'silver_finished_products'. A Select transformation retained only the seven ERD defined

columns, removing the raw 'finished_product' name column. The processed data was written to the 'silver_sales_orders' dataset.



Figure 5.2.3.9 df_silver_sales_orders

df_silver_bill_of_materials

Figure 5.2.3.10 shows the 'silver_bill_of_materials' table is a derived table that does not have a direct source in the raw data. Instead, it is computed from two Silver layer tables 'silver_production_orders' and 'silver_sales_orders'.

The dataflow begins with two independent source streams. The first stream reads from 'silver_production_orders' and applies an Aggregate transformation grouping by both 'finished_product_id' and 'material_id', computing the total material consumption as 'sum(quantity_consumed)' aliased as 'total_consumed'. The second stream reads from 'silver_sales_orders', applies a Filter transformation to retain only records where 'status' == "delivered" and then applies an Aggregate transformation grouping by 'finished_product_id' to compute the total delivered quantity as 'sum(quantity_ordered)' aliased as 'total_delivered'.

A Left Join transformation was applied between the two aggregated streams on the condition 'totalConsumed@finished_product_id==totalDelivered@finished_product_id', retaining all material consumption records even if no matching delivered sales exist. A Derived Column transformation then calculated the 'ratio_required' column using the expression 'round(toFloat(total_consumed) toFloat(total_delivered), 2)', representing the quantity of each raw material required to produce one unit of the corresponding finished product. A Filter transformation removed rows where 'ratio_required' was null or where 'total_delivered' was zero to avoid division errors. Finally, a Select transformation retained only the three ERD defined columns 'finished_product_id', 'material_id' and 'ratio_required' before writing the output to the 'silver_bill_of_materials' dataset.



Figure 5.2.3.10 df_silver_bill_of_materials

5.3.3 Gold Layer

The Gold Layer is the final stage of the data pipeline, where the cleaned and standardized data from the Silver Layer is transformed into business-ready datasets. Unlike the Silver Layer, which focuses on data cleaning, validation, and standardization, the Gold Layer focuses on applying business rules, preparing analytical tables, and generating datasets that can be used for reporting and dashboard visualization.

In this project, the Gold Layer was implemented using Azure Data Factory data flows. The Gold Layer pipeline consists of several data flows that generate both dimension and fact tables.

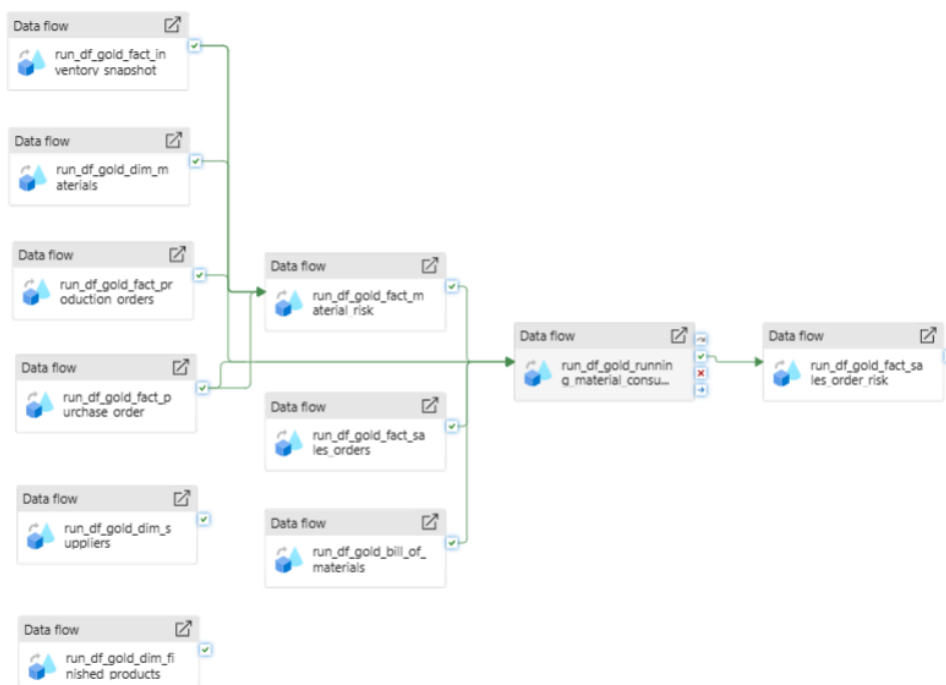


Figure 5.3.3.1 Gold layer pipeline

Figure 5.3.3.1 shows the overall Gold Layer pipeline in Azure Data Factory. The pipeline executes multiple Gold Layer data flows and connects the outputs into later analytical

processes. The basic dimension and fact tables are generated first, followed by material risk analysis, running material consumption, and sales order risk calculation. This shows that the Gold Layer is not only storing cleaned data, but also preparing business-level datasets for inventory risk management.

df_gold_bill_of_materials

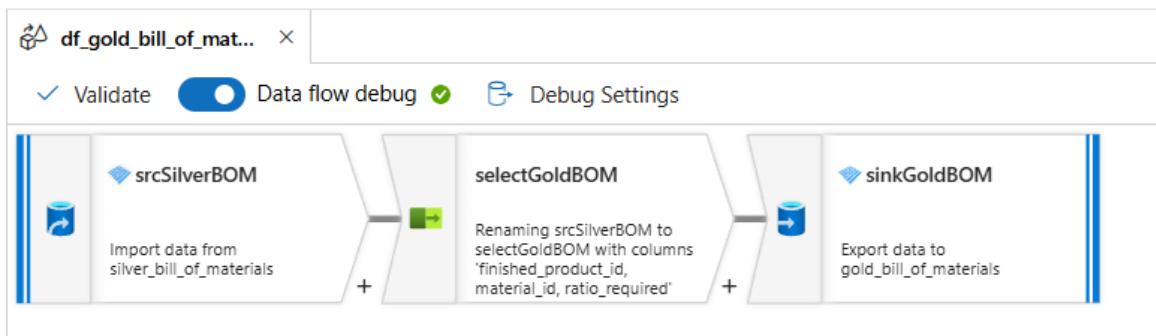


Figure 5.3.3.2 df_gold_bill_of_materials

The df_gold_bill_of_materials data flow creates the Gold Layer bill of materials dataset. It imports data from silver_bill_of_materials, selects important fields such as finished product ID, material ID, and ratio required, and exports the output into gold_bill_of_materials. The gold_bill_of_materials table is later used in the running material consumption data flow to calculate material demand from sales orders.

df_gold_dim_suppliers

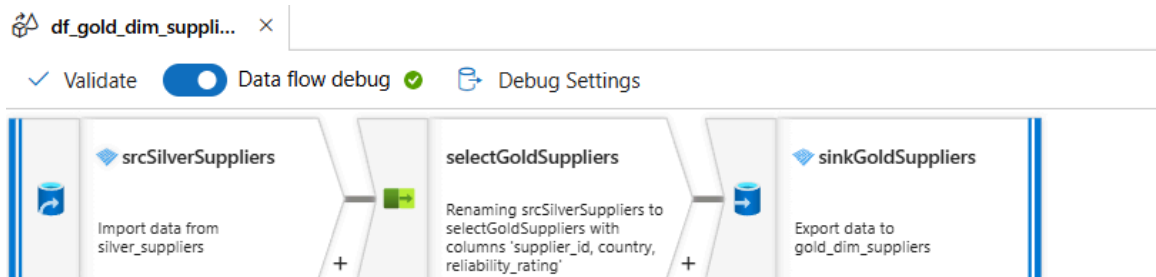


Figure 5.3.3.2 df_gold_dim_suppliers

The df_gold_dim_suppliers data flow creates the Gold Layer supplier dimension table. It imports data from silver_suppliers, selects the required supplier-related columns, and exports the output into gold_dim_suppliers. This data flow prepares supplier information such as supplier ID, country, and reliability rating. The output helps the system connect purchase order records with supplier details.

df_gold_dim_finished_products

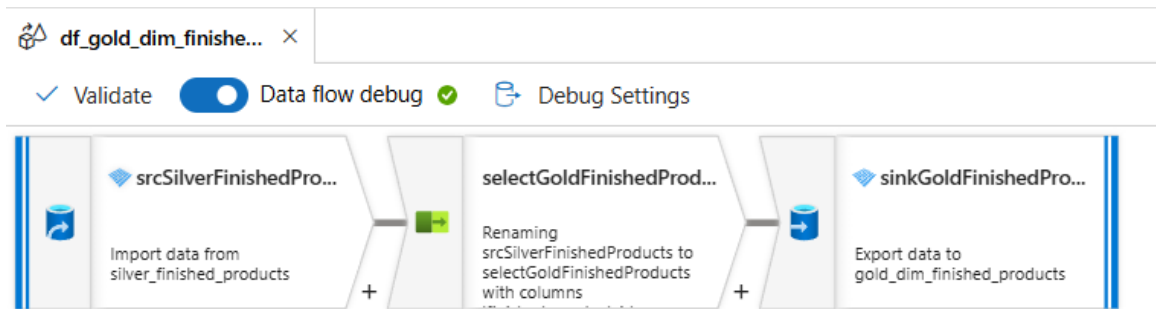


Figure 5.3.3.3 df_gold_dim_finished_products

The df_gold_dim_finished_products data flow creates the Gold Layer finished product dimension table. It imports data from silver_finished_products, selects the required finished product columns, and exports the output into gold_dim_finished_products. This table stores descriptive information about finished products. It is used together with the bill of materials dataset to identify which raw materials are required to produce each finished product.

df_gold_fact_inventory_snapshot

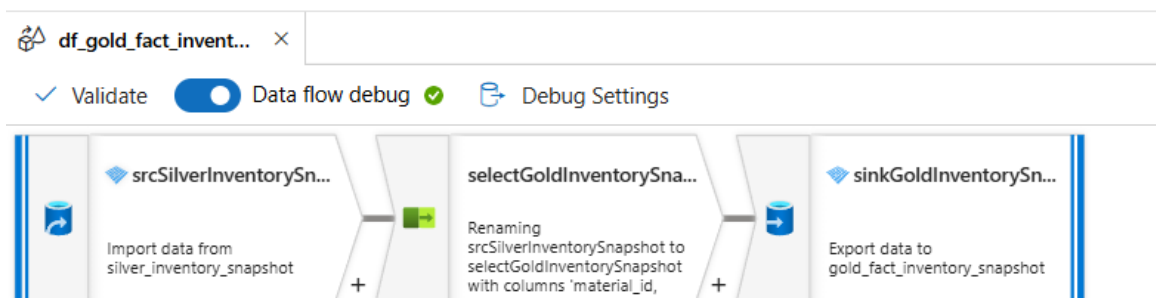


Figure 5.3.3.4 df_gold_fact_inventory_snapshot

The df_gold_fact_inventory_snapshot data flow creates the Gold Layer inventory snapshot fact table. It imports data from silver_inventory_snapshot, selects the required inventory columns, and exports the output into gold_fact_inventory_snapshot. This fact table stores inventory status information, such as material quantity, stock date, and inventory-related values. It provides the latest available inventory information for each material.

df_gold_fact_purchase_order

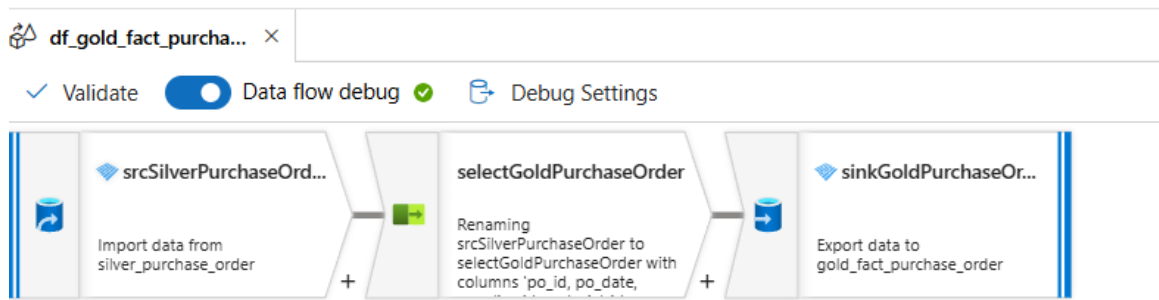


Figure 5.3.3.5 df_gold_fact_purchase_order

The df_gold_fact_purchase_order data flow creates the Gold Layer purchase order fact table. It imports data from silver_purchase_order, selects important purchase order columns such as purchase order ID, purchase order date, supplier ID, material ID, and delivery-related information, and exports the output into gold_fact_purchase_order. This dataset is used to track incoming materials from suppliers. It helps the system identify whether purchase orders are still outstanding and whether material arrival may reduce stockout risk.

df_gold_fact_production_orders

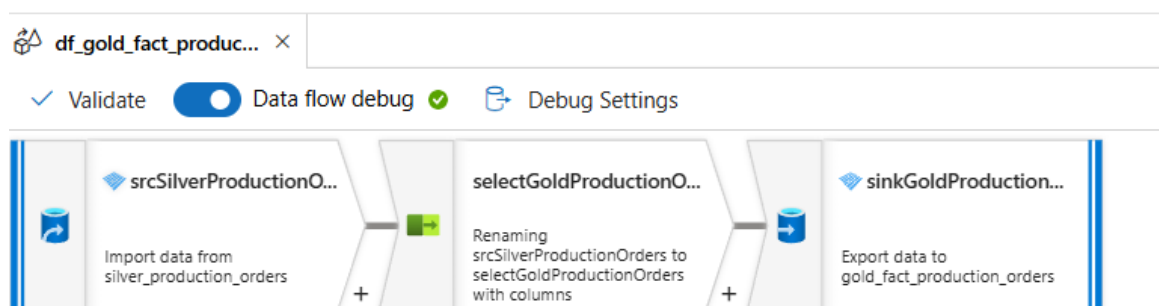


Figure 5.3.3.6 df_gold_fact_production_orders

The df_gold_fact_production_orders data flow creates the Gold Layer production order fact table. It imports data from silver_production_orders, selects the required production related columns, and exports the output into gold_fact_production_orders. This dataset stores production activity information. It is used to understand material consumption from production records.

df_gold_fact_sales_orders

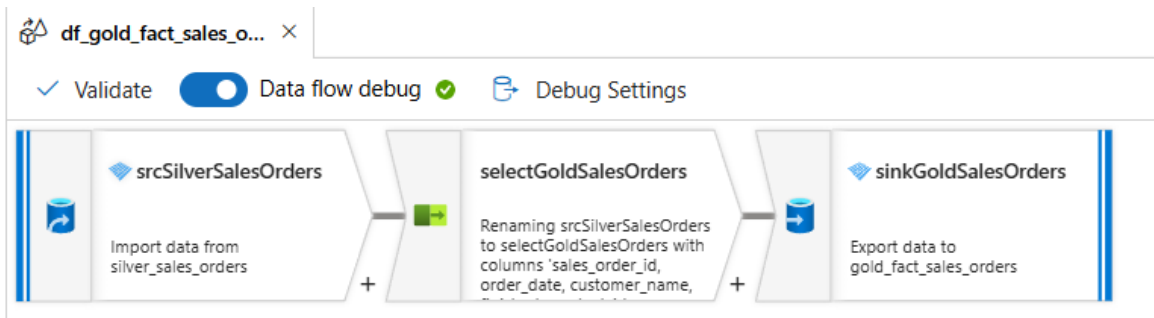


Figure 5.3.3.7 df_gold_fact_sales_orders

The df_gold_fact_sales_orders data flow creates the Gold Layer sales order fact table. It imports data from silver_sales_orders, selects important sales order fields such as sales order ID, order date, customer name, finished product ID, and required delivery date, and exports the output into gold_fact_sales_orders. This dataset stores customer demand information. It is important because sales orders create demand for finished products, which then creates demand for raw materials through the bill of materials.

df_gold_fact_material_risk

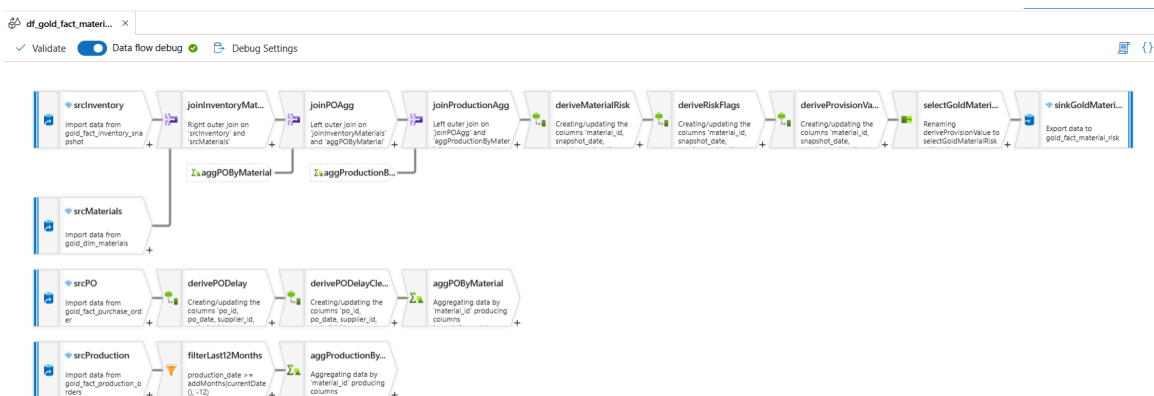


Figure 5.3.3.8 df_gold_fact_material_risk

The df_gold_fact_material_risk data flow creates the main material risk analysis dataset. It combines data from gold_fact_inventory_snapshot, gold_dim_materials, gold_fact_purchase_order, and gold_fact_production_orders. This data flow performs several important steps. First, inventory data is joined with material details to provide descriptive information for each material. Then, purchase order data is aggregated by material to identify incoming supply. Production order data is filtered for the latest 12 months and aggregated by material to calculate recent material consumption. After that, derived columns are created to generate material risk indicators. These indicators may include Recoverable Asset status, Magna Carta dead stock status, expired inventory

status, risk flags, and provision value. The final output is exported into gold_fact_material_risk.

df_gold_running_material_consumption

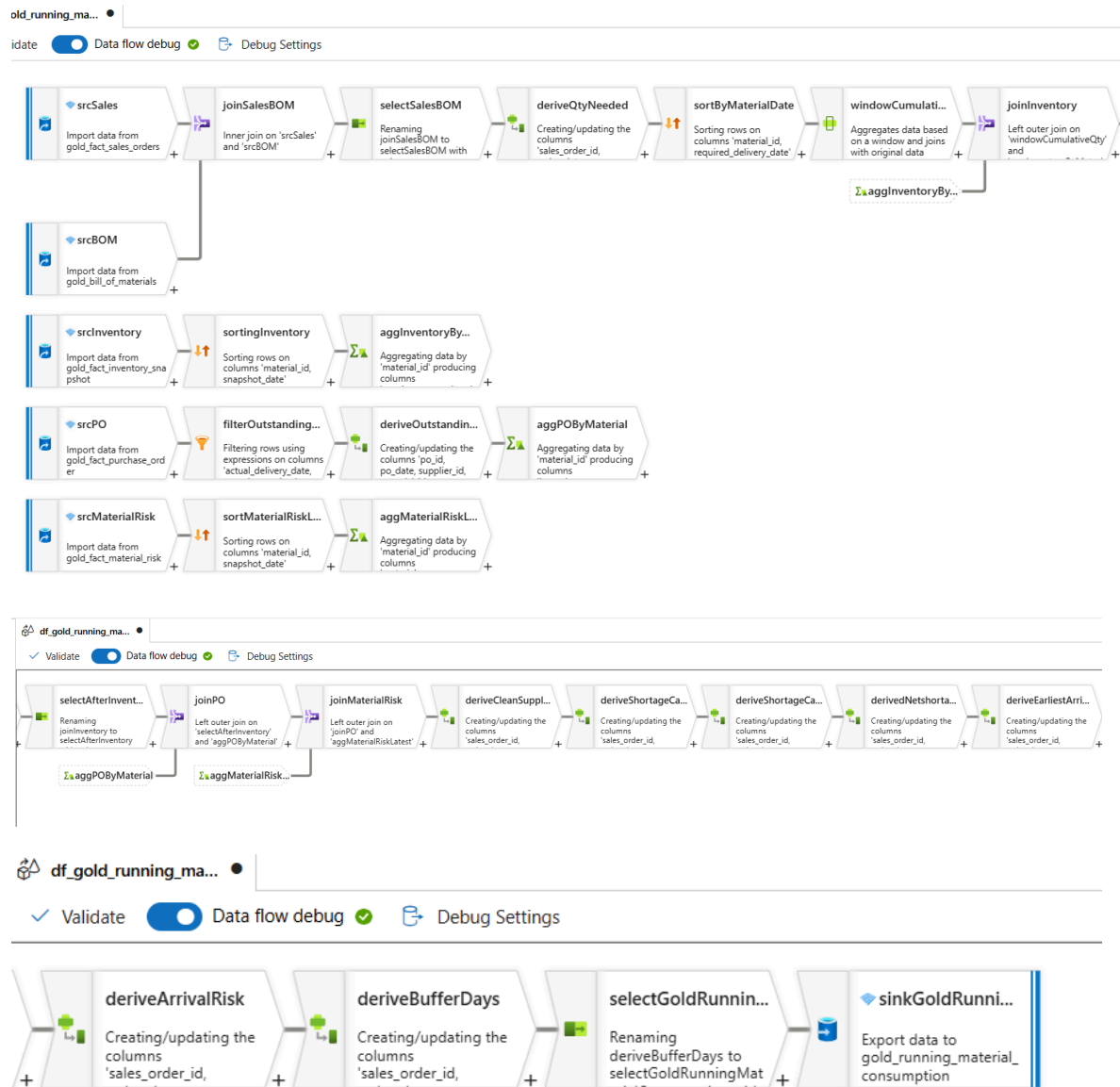


Figure 5.3.3.9 df_gold_running_material_consumption

The df_gold_running_material_consumption data flow calculates material demand and shortage risk based on sales orders, bill of materials, inventory, purchase orders, and material risk data. This data flow starts by importing data from gold_fact_sales_orders and gold_bill_of_materials. These two datasets are joined to calculate the quantity of raw materials required for each sales order. The data is then sorted by material and required delivery date to calculate cumulative material requirements. Next, the result is joined with inventory data to compare required material quantity with available stock. Purchase order

information is also joined to consider incoming supply, while material risk information is joined to include existing material risk conditions. Derived columns are then created to calculate shortage-related indicators such as clean supply quantity, shortage quantity, net shortage, earliest arrival information, arrival risk, and buffer days. The final output is exported into gold_running_material_consumption.

df_gold_fact_sales_order_risk



Figure 5.3.3.10 df_gold_fact_sales_order_risk

The df_gold_fact_sales_order_risk data flow creates the final sales order risk dataset. It imports data from gold_running_material_consumption, derives blocked material flags, aggregates the result by sales order ID, and creates sales order risk indicators. This data flow converts material-level risk information into sales order-level risk information. For example, if one or more required materials are insufficient, delayed, or blocked, the related sales order can be marked as risky. The final output is exported into gold_fact_sales_order_risk. This dataset is useful for identifying which sales orders may be affected by material shortages, blocked materials, or late material arrivals.

5.3.4 Azure Synapse Analytics

Workspace Setup & Data Lake Integration

The Synapse Analytics workspace synapseppg was created in the Azure Data Lake Storage Gen2 account named ppgdatalake01 directly as its primary storage, which means that all data for the pipeline will reside in the same Azure environment. The gold container was then deployed and checked that all 11 gold layer folders were populated by the ADF pipelines in Synapse Studio. A database named gold_db was created with the Built-in Serverless SQL Pool for the analytics database as shown in Figure 5.3.4.1, and each gold layer parquet folder contained in the data lake was mapped to a corresponding view in the analytics database. All this without unnecessarily storing data, without the need for a dedicated always-on compute cluster, and without having to duplicate data for structured SQL querying.

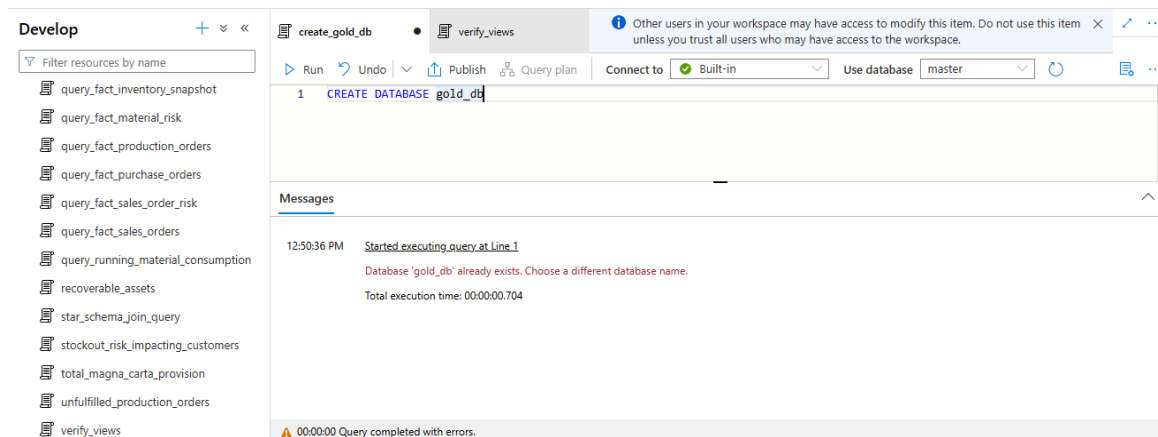


Figure 5.3.4.1: Create gold_db

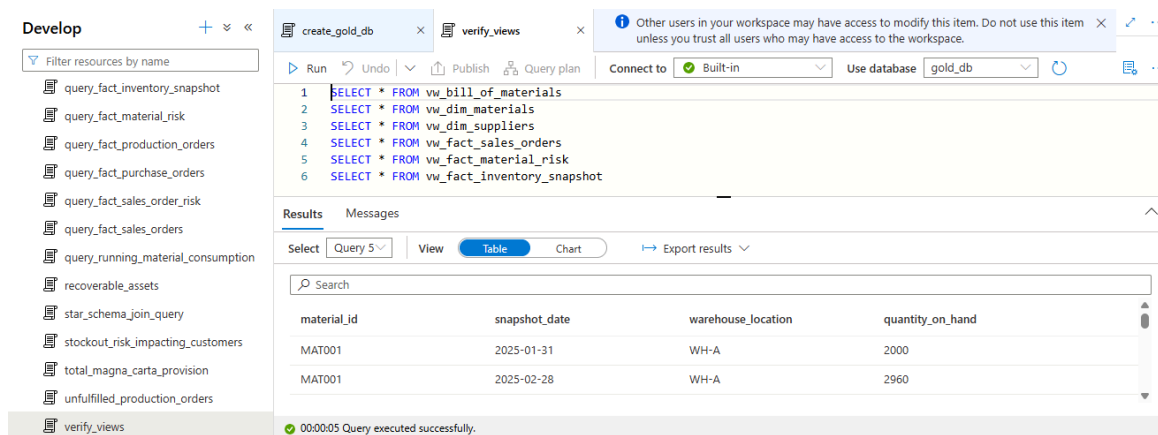
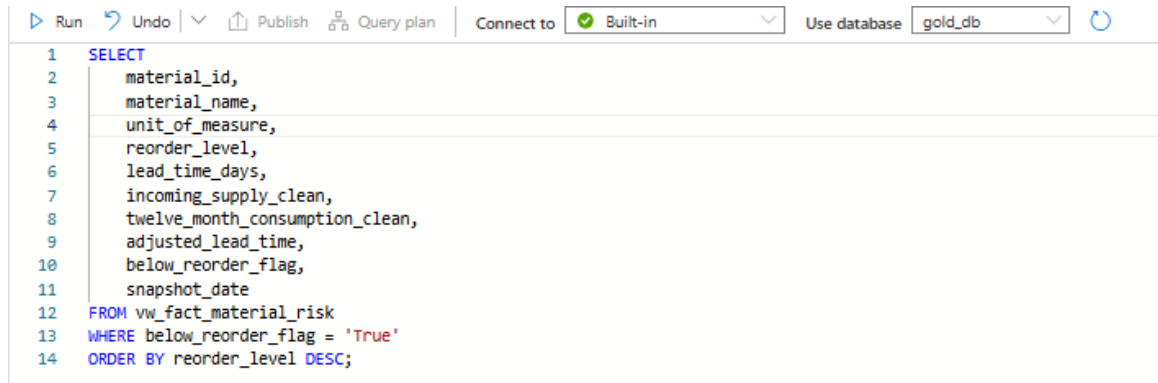


Figure 5.3.4.2: Verify Views

Analytics Queries

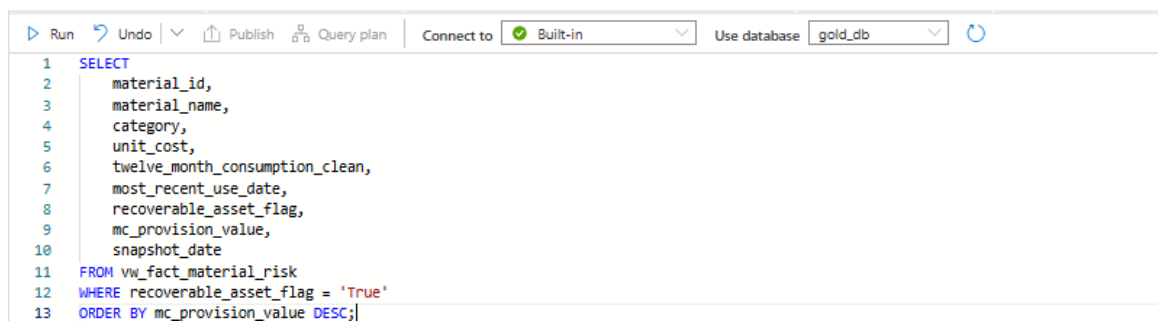
The seven business analytics queries were designed within gold_db to satisfy the business analytics requirements from the project dashboard as specified in the problem statement. The vw_fact_material_risk view was used as the primary data source since it had the most complete set of analytical columns available such as reorder_level, below_reorder_flag, mc_dead_stock_flag, recoverable_asset_flag, mc_expired_flag, and mc_provision_value, which are pre-computed by the gold layer ADF flows. Figure 5.3.4.3 was filtered and critical materials at zero stock were identified such as MAT018 (Plastic Drum 20L), MAT013 (Mineral Spirits), MAT008 (Epoxy Resin) with below reorder flag = True and reorder range from 400-600 units. The flag columns were also used in Figure 5.3.5.4 to mark assets that were recoverable, and in Figure 5.3.4.5 to mark the dead stock material, and the third one, in Figure 5.3.4.7, combined mc_provision_value to arrive at the financial write-down amount that would be required. Figure 5.3.4.8 joined vw_fact_production_orders with vw_fact_inventory_snapshot on material_id, using CASE statement to set an unfulfilled flag on those production orders that had

quantity_on_hand less than quantity_consumed, including PRD-083 that had a quantity_consumed of 320 units of material MAT013 and quantity_on_hand of 0. However, the results sets were empty for Figure 5.3.4.6 and Figure 5.3.4.9, both of which were reported as valid and meaningful analyses that identified healthy stock expiration and no active disruptions to sales orders, respectively.



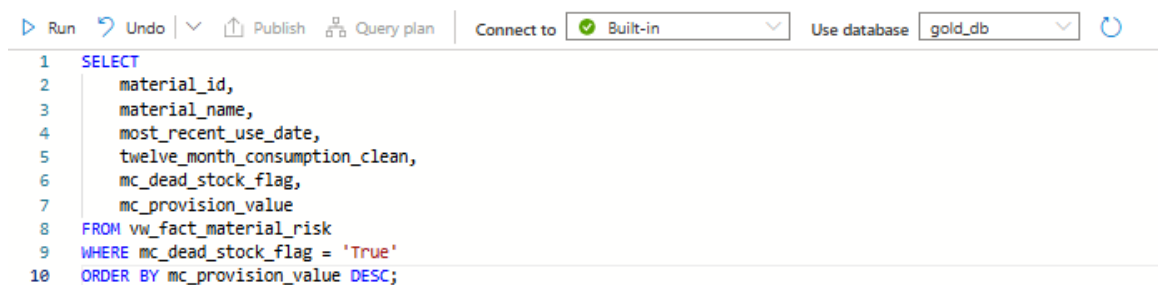
```
1 SELECT
2     material_id,
3     material_name,
4     unit_of_measure,
5     reorder_level,
6     lead_time_days,
7     incoming_supply_clean,
8     twelve_month_consumption_clean,
9     adjusted_lead_time,
10    below_reorder_flag,
11    snapshot_date
12 FROM vw_fact_material_risk
13 WHERE below_reorder_flag = 'True'
14 ORDER BY reorder_level DESC;
```

Figure 5.3.4.3: Materials Below Reorder Level



```
1 SELECT
2     material_id,
3     material_name,
4     category,
5     unit_cost,
6     twelve_month_consumption_clean,
7     most_recent_use_date,
8     recoverable_asset_flag,
9     mc_provision_value,
10    snapshot_date
11 FROM vw_fact_material_risk
12 WHERE recoverable_asset_flag = 'True'
13 ORDER BY mc_provision_value DESC;
```

Figure 5.3.4.4: Recoverable Assets



```
1 SELECT
2     material_id,
3     material_name,
4     most_recent_use_date,
5     twelve_month_consumption_clean,
6     mc_dead_stock_flag,
7     mc_provision_value
8 FROM vw_fact_material_risk
9 WHERE mc_dead_stock_flag = 'True'
10 ORDER BY mc_provision_value DESC;
```

Figure 5.3.4.5: Magna Carta Dead Stock

```

1 SELECT
2     material_id,
3     material_name,
4     most_recent_use_date,
5     twelve_month_consumption_clean,
6     mc_expired_flag,
7     mc_provision_value,
8     recoverable_asset_flag
9 FROM vw_fact_material_risk
10 WHERE mc_expired_flag = 'True'
11 ORDER BY mc_provision_value DESC;

```

Figure 5.3.4.6: Magna Carta Expired

```

1 SELECT
2     SUM(mc_provision_value) AS total_provision_value,
3     COUNT(material_id) AS total_materials_affected
4 FROM vw_fact_material_risk
5 WHERE mc_provision_value > 0;

```

Figure 5.3.4.7: Total Magna Carta Provision

```

1 SELECT TOP 5 * FROM vw_fact_production_orders

```

Figure 5.3.4.8: Unfulfilled Production Orders

```

1 SELECT s.sales_order_id, s.customer_name, r.has_blocked_material, r.sales_order_delay_flag
2 FROM vw_fact_sales_order_risk r
3 JOIN vw_fact_sales_orders s ON r.sales_order_id = s.sales_order_id
4 WHERE r.sales_order_delay_flag = 'True'
5 ORDER BY s.customer_name

```

Figure 5.3.4.9: Stockout Risk Impacting Customers

Galaxy Schema Design

All the fact and dimension tables were imported directly from the gold container in ppgdatelake01 to the Lake Database created in Synapse Studio to formally model and document the analytical data structure. The schema is based on a galaxy schema, Figure 5.3.4.11 with 7 fact tables in the center including Fact_material_risk, Fact_inventory_snapshot, Fact_production_orders, Fact_purchase_order, Fact_sales_orders, Fact_sales_order_risk and Running_material_consumption, each containing measurable facts in inventory, procurement, production, and sales contexts. Three dimension tables including Dim_materials, Dim_suppliers, Dim_finished_products are associated with these, giving descriptive context to the facts, and a Bill_of_materials table is a reference table linking products to the materials they require. Formal

relationships were created between the tables using the common key columns as a basis for establishing relationships, such as `material_id` is used to establish a relationship between Fact tables and `Dim_materials`, `supplier_id` is used to establish a relationship between `Fact_purchase_order` and `Dim_suppliers`, and so on to generate a visual relationship diagram in Synapse Studio. Furthermore, a SQL join query in Figure 5.3.4.10 was prepared in order to validate these relationships programmatically, and it was successfully able to return attributes combined from `vw_fact_material_risk`, `vw_dim_materials`, `vw_fact_purchase_order` and `vw_dim_suppliers` in a single result set.

```

1  SELECT
2      f.material_id,
3      m.material_name,
4      m.category,
5      m.unit_of_measure,
6      p.supplier_id,
7      s.country,
8      s.reliability_rating,
9      i.quantity_on_hand,
10     m.reorder_level,
11     f.below_reorder_flag,
12     f.mc_dead_stock_flag,
13     f.recoverable_asset_flag,
14     f.mc_provision_value,
15     f.snapshot_date
16 FROM vw_fact_material_risk      f
17 LEFT JOIN vw_dim_materials    m ON f.material_id = m.material_id
18 LEFT JOIN vw_fact_purchase_order p ON f.material_id = p.material_id
19 LEFT JOIN vw_dim_suppliers    s ON p.supplier_id = s.supplier_id
20 LEFT JOIN vw_fact_inventory_snapshot i ON f.material_id = i.material_id
21                                     AND f.snapshot_date = i.snapshot_date
22 ORDER BY f.material_id;

```

Figure 5.3.4.10: Galaxy Schema Join Query

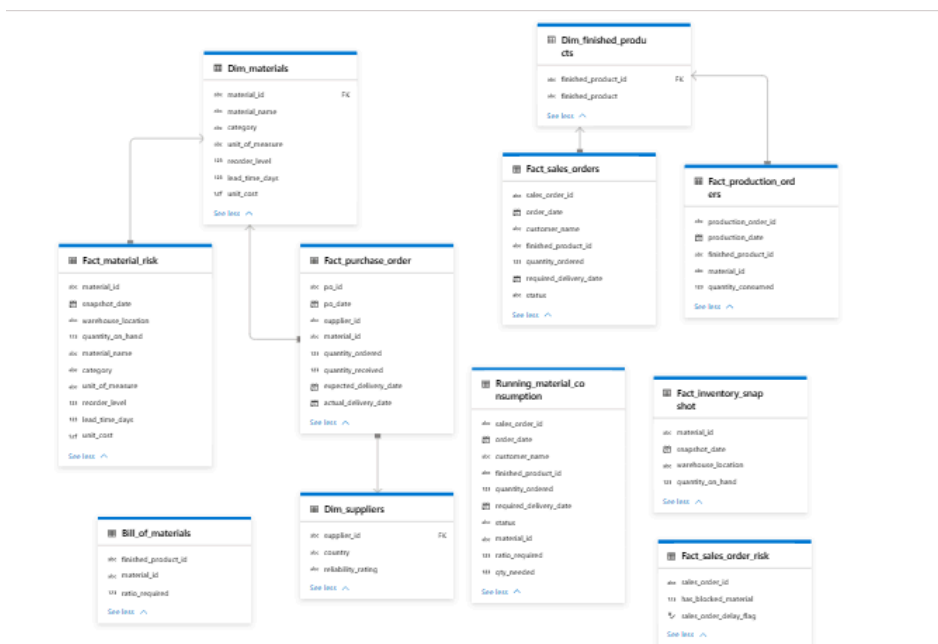


Figure 5.3.4.11: Galaxy Scheme Design

5.4 Dashboard and Interface Development

The data was retrieved from Azure Synapse via the Serverless SQL endpoint. All views created in Synapse were loaded into PowerBI.

The relationships between the tables were established as shown in Figure 5.4.1.



Figure 5.4.1: Model View in PowerBI

Several DAX measures and calculated columns were created to facilitate dashboard building. These include measures for dead stock count, expired material count, expired material value, recoverable assets count, recoverable value, and total material count.

Three dashboards were developed with each serving a different purpose. The executive overview in Figure 5.4.2 provides a higher level of risk visibility. The inventory analysis dashboard in Figure 5.4.3 dives deeper to see the root cause of the risks, while the sales impact dashboard shows the sales order impact. All of the three dashboards were separated into three sections, section A, B and C, with some variation. Section A mostly

shows the numeric summary of the interested information. Section B displays aggregated results in the form of graphs and Section C includes the details in the form of tables.

In the executive overview dashboard in Figure 5.4.2.

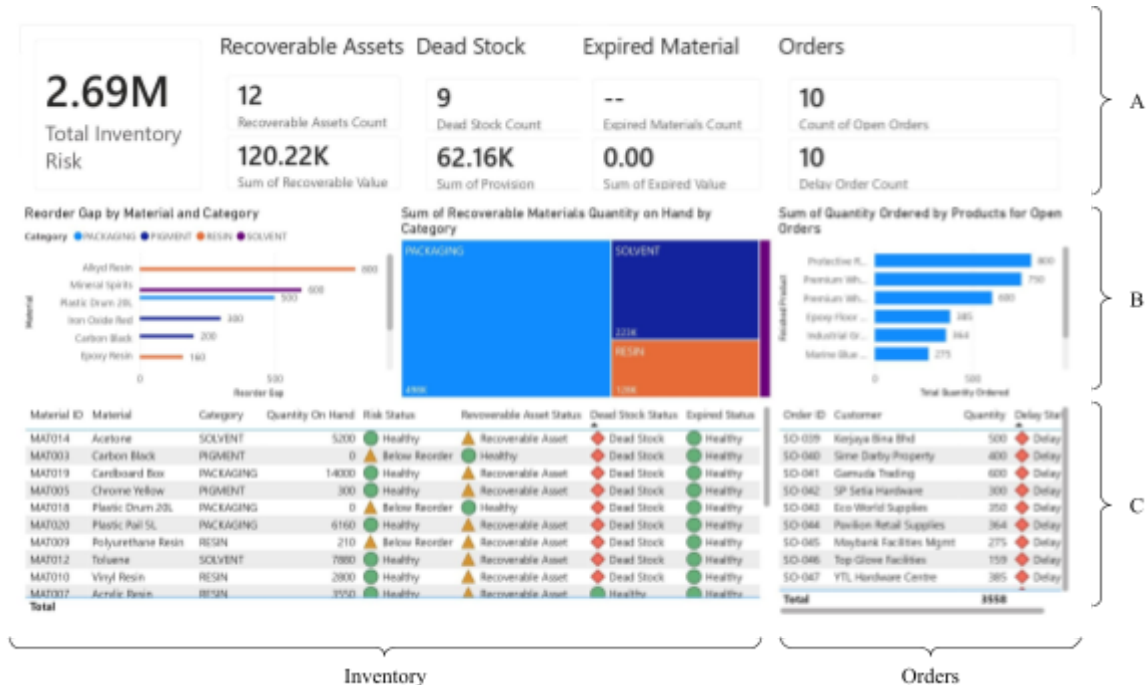


Figure 5.4.2: Executive Overview Dashboard

In section A, scorecards were used to represent total inventory risk, number of materials flagged as recoverable assets, dead stock and expired materials, and associated values of these flagged materials. The dashboard also includes scorecards showing the number of open orders and delayed orders. In section B, bar charts and tree maps provide a per-material, per-category and per-product overview. Finally in Section 2, two tables were used to display the status of the materials and sales orders.

In the inventory analysis dashboard, as shown in figure 5.4.3, provides a detailed view of inventory.

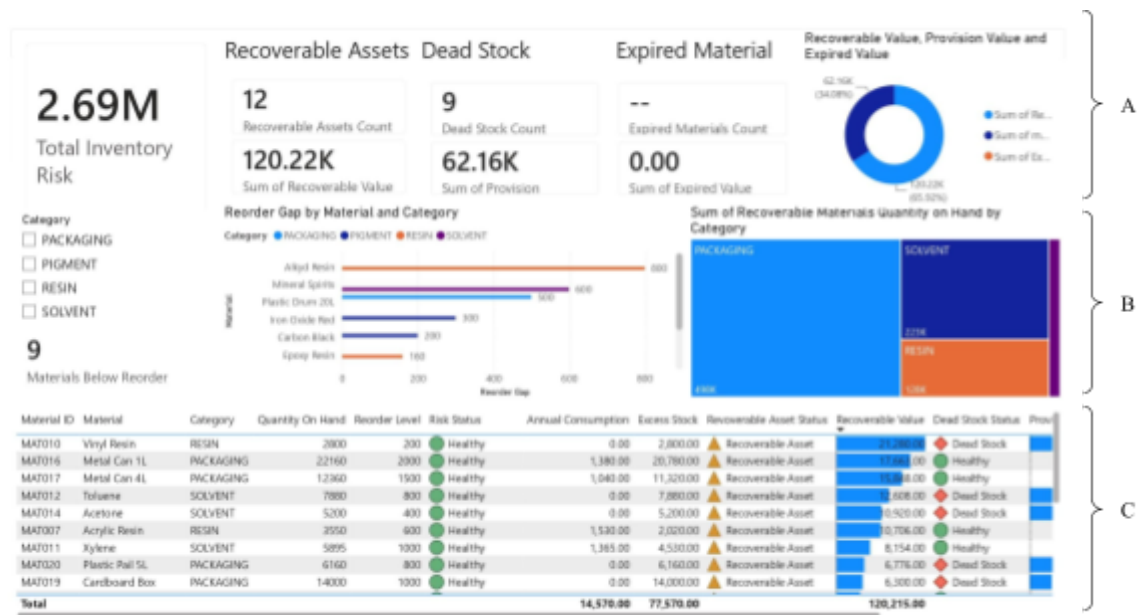


Figure 5.4.3: Inventory Analysis Dashboard

In addition to the existing visuals from the inventory section of executive overview dashboard, the number of materials below reorder level was also added in the form of scorecards and a donut visual was used to display the proportion of the recoverable value, dead stock value and expired material value. In section C, other relevant per material information were included including reorder levels, annual consumption, excess stock, recoverable value, provision value and expired value. This dashboard enables planners to investigate the causes of inventory risks and prioritise actions.

The final dashboard includes a sales impact dashboard as shown in figure 5.4.4.

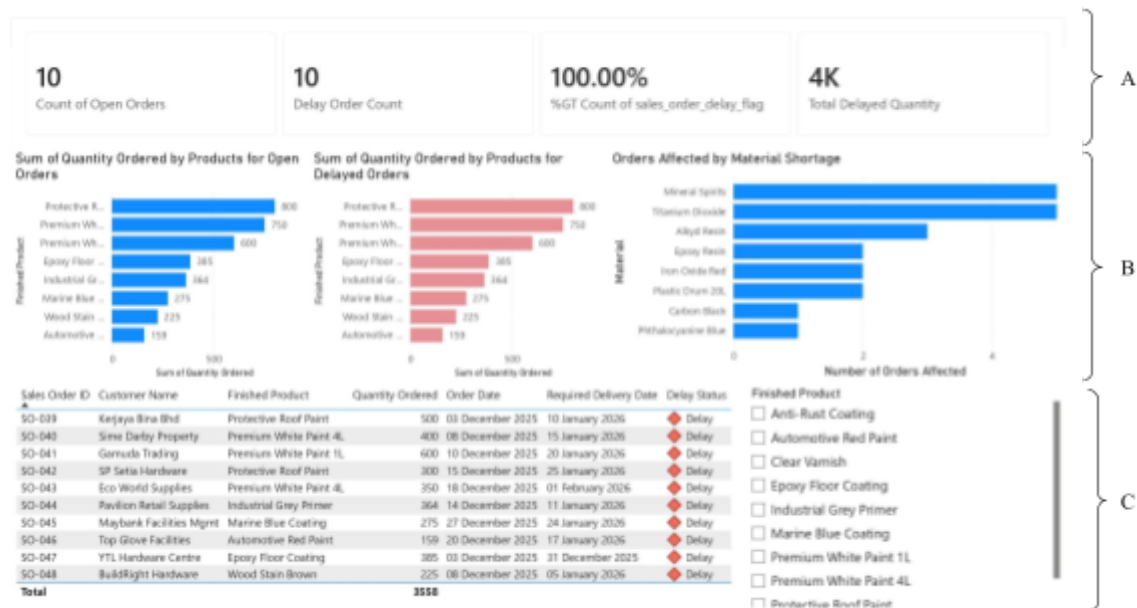


Figure 5.4.4: Sales Impact Dashboard

This dashboard shows the percentage of delay orders and total delayed quantity of ordered products, in addition to the number of sales orders and the number of delayed sales orders. Besides, it includes three horizontal bar charts that show the number of finished products in both open and delayed orders. This dashboard highlights the operational impact of inventory risks on order fulfillment.

5.5 Testing

Testing was conducted to ensure the data pipeline operates accurately and reliably across all stages. In this project, black-box testing and white-box testing were used. Black-box testing validates the system's functionality by checking whether correct outputs are produced from given inputs, without considering internal logic, whereas white-box testing focuses on verifying the internal processes, including data transformations and workflow logic within the pipeline. Together, these approaches ensure both functional correctness and internal reliability of the system.

5.5.1 Black Box Testing

Black-box testing was conducted to validate the functionality of the data pipeline without examining the internal implementation details. The testing focused on verifying whether the system produces the expected outputs when given specific inputs, ensuring that each stage of the pipeline operates correctly from a user and business perspective. In this project, black-box testing was applied across the end-to-end data pipeline, including data ingestion, transformation, storage, and visualization.

At the ingestion stage, testing focused on verifying that all six source datasets were correctly extracted and loaded into the Azure Data Lake Bronze layer without data loss or corruption. In addition, basic data quality rules were validated to ensure that null values were detected, data types were correctly assigned, and duplicate records were appropriately handled. For the Silver layer, testing was carried out to ensure that data cleaning and processing logic produced consistent and accurate outputs. This included verifying that inconsistencies were removed and that datasets were properly structured for analytical use. At the Gold Layer, black-box testing ensured that business rules such as inventory calculations, expired material identification, and supply risk indicators were correctly generated. Furthermore, Azure Synapse Analytics was tested to confirm that queries executed efficiently and returned accurate results based on the processed datasets. Finally, Microsoft Power BI dashboards were validated to ensure that visualizations correctly reflected the underlying data and provided accurate insights for decision-making.

In conclusion, black-box testing confirmed that the data pipeline functions reliably end-to-end, producing accurate, consistent, and meaningful outputs across all stages.

5.5.2 White Box Testing

We conducted white box testing to verify that the internal operations of the data pipeline were mathematically and logically sound across every stage. This process targeted the Silver layer dataflows, Gold layer logical rules, and Azure Synapse Analytics query scripts.

During Silver layer validation, testing verified that string functions like `trim()` and `upper()` normalized text, while numeric functions like `toFloat()` and `round()` handled data type conversions and rounding. Data cleaning checks, including null primary key removal, deduplication using `first()`, and foreign key validations, prevented corrupted records from reaching the output. Additionally, the `crc32` hashing function generated consistent surrogate keys for matching product names, and the bill of materials calculation omitted records with zero delivered quantity to prevent division errors.

For the Gold layer, we evaluated the conditional logic that generates risk metrics. The `recoverable_asset_flag` was verified to return `True` only when available stock exceeds twelve month consumption. Boundary tests confirmed that the `mc_dead_stock_flag` triggers precisely at nine months of zero usage, with the `mc_provision_value` calculating

the correct financial write down value. Testing also confirmed that the material_risk attribute assigned the BLOCKED, TIGHT, and OK classifications correctly, while the sales_order_delay_flag successfully promoted the BLOCKED status to the order header.

In Synapse Analytics, we confirmed that left join operations preserved all material records when matching purchase orders were absent and key joins correctly mapped supplier details. Furthermore, empty datasets from queries tracking expired stock or inventory depletion risks reflected actual business realities rather than pipeline errors.

In summary, this testing demonstrated that the pipeline functions correctly at every stage, ensuring accurate risk assessments and dependable datasets for Power BI reporting.

5.6 Security and Data Privacy

Security

1. Identity and Access Management

All team members login with their institutional UTM Microsoft accounts on all the services including Azure Data Factory (ADF), Databricks, and Synapse. Azure Role Based Access Control (RBAC) limits access to the resource group ppg-project-rg, and roles are clearly defined, with the project admin having Owner access and all team members having Contributor access. This will allow members with permission to interact with, alter, or activate any segment of the pipeline.

2. Network Security

All ADF pipelines rely on AutoResolveIntegrationRuntime, a Microsoft managed runtime which will run entirely inside of Azure's private cloud network, so that no data will traverse the public internet whilst the pipeline is running. The storage account ppgdatalake01 has been configured to deny public access, allowing only authenticated access by Azure identities.

3. Create, and Manage Identities and Credentials

In all services, password, connection strings or storage keys are not stored anywhere in pipeline code or configuration, and all services are authenticated with each other using a managed identity assigned to the service, at service level. The data lake is authenticated by ADF's Managed Identity and Synapse is authenticated by its own Managed Identity to the gold container.

4. Pipeline Access Control

Only contributor-level users can trigger, modify or publish ADF pipelines, and all pipeline runs are recorded on the ADF Monitor tab, allowing for a complete audit trail of all data movements. In Synapse, a special SQL admin account was created that requires strong password requirements to access SQL Pools, in addition to the Azure Portal authentication. All analytics queries go through gold_db views and not directly to the raw parquet files, providing an abstraction layer that protects the underlying storage paths from end users.

Data Privacy

1. Data Layer Separation

The Medallion architecture ensures natural data privacy boundaries with three separate containers which are bronze, silver, and gold, each with their own access permissions. The raw data of transactions remains in the bronze layer and is only available for access via ADF, whereas the aggregated and analytics-ready data is exposed to Synapse for queries in the gold layer. This ensures that the raw data is not sent to end users and reporting tools directly.

2. Data Minimization

At the bronze ingestion stage, ADF data flows perform filter and aggregate transformations, which ensure that only the required fields and legitimate records are carried along in the pipeline. Further, invalid, null or duplicate records are eliminated from the exposed data via the validation data flows, thus minimizing unnecessary data exposure. The data will be aggregated at the business level and stripped of any personally identifiable information by the time it reaches the gold layer.

3. Data Access Control

The access permissions of each container is independent, and they are granted by the least privilege principle. An additional layer of access control, end users access data only via gold_db views in Synapse and not the raw storage files. The underlying parquet files in the data lake cannot be accessed directly, except through the managed service identities.

4. Audit and Monitoring

Every pipeline run that crosses bronze, silver, and gold layers is recorded on the ADF monitor tab, and that includes action time, status, and data flow information, giving a complete audit trail of all data flows. All SQL query executions are logged in Synapse, showing who accessed what data and when. The Azure Activity Log is populated with all resource-level actions for ppg-project-rg, making it easy to track any changes to access, configuration, and resources.

5. Institutional Data Governance

The project is hosted in the Azure for Students subscription, which includes built-in spending limits, and limited permissions to ensure that resources are not improperly deployed and points are not accidentally exposed. Project resources are only available to those who have a project within institutional email, so all project data is contained within the university's Microsoft tenant and governance.

5.7 Summary

This chapter presented the implementation and testing of the Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The system was developed using Microsoft Azure services and Power BI, where Azure Data Factory was used to orchestrate the data pipeline, while Azure Data Lake Storage Gen2 was used to store data across the Bronze, Silver, and Gold layers. Azure Synapse Analytics was used for querying analytics-ready datasets, and Power BI was used to create interactive dashboards for inventory monitoring. The implementation followed the Medallion Architecture. In the Bronze layer, six raw CSV datasets were ingested and stored to preserve the original data for traceability. In the Silver layer, the data was cleaned, standardized, validated, and converted into structured Parquet format. In the Gold layer, business rules were applied to produce analytics-ready datasets for identifying Recoverable Assets, Magna Carta dead stock, expired inventory, stockout risks, and financial risk values.

Testing was also carried out to ensure that the system functions correctly and meets the project requirements. Black box testing was used to verify that the system outputs matched the expected results, while white box testing was used to check the internal logic, data transformations, and business rules. Overall, the implementation and testing confirmed that the developed system is able to process inventory data, generate reliable analytical outputs, and support better decision-making through dashboard-based inventory risk monitoring.

CHAPTER 6

CONCLUSION

6.1 Introduction

This chapter presents the conclusion of the project titled “PPG Recoverable Assets (RA) & Inventory Risk Management (IRM)”, developed in collaboration with PPG Coatings (Malaysia) Sdn. Bhd. The chapter summarises the overall achievements of the project, evaluates the extent to which the project objectives were met, identifies the key constraints encountered during system development, and proposes directions for future improvement.

Effective inventory management is a critical operational challenge for manufacturing organisations. Poor visibility into inventory status can result in excess stock accumulation, dead stock, expired materials, and stockout risks that lead to significant financial losses and disruptions in fulfilling customer orders. PPG Coatings (Malaysia) Sdn. Bhd. faced these challenges due to fragmented data sources that made it difficult to obtain a consolidated and reliable view of inventory conditions. This project was developed to address these issues by designing and implementing a structured, end-to-end data engineering solution that integrates multiple operational datasets, applies inventory risk business rules, and delivers actionable insights through interactive dashboards.

The project was guided by four key objectives: first, to analyse the inventory risk management problem domain and evaluate existing data engineering approaches for detecting Recoverable Assets (RA), dead stock, and stockout risks; second, to design a three-layer Medallion Architecture pipeline consisting of Bronze, Silver, and Gold layers for data integration and business rule application; third, to implement the proposed pipeline on Microsoft Azure using Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, and Azure Synapse Analytics; and fourth, to develop and evaluate an interactive dashboard using Microsoft Power BI that visualises inventory risks and supports data-driven decision-making.

The significance of this project lies in its contribution of a practical, scalable data engineering solution that improves inventory transparency, strengthens risk detection, and supports proactive inventory management at PPG Coatings (Malaysia) Sdn. Bhd. By converting raw, fragmented operational data into structured and meaningful business insights, the system enables inventory managers and supply chain managers to shift from reactive tracking to proactive risk mitigation, ultimately reducing operational costs and

improving decision-making efficiency.

6.2 System Contribution and Achievement

This section discusses the contributions and achievements of the project. It describes the main findings from the literature review and evaluates the achievement of the project objectives. The system's overall contribution is evaluated by its ability to support inventory risk analysis, improve data accessibility and provide meaningful insights through an end-to-end data engineering pipeline and interactive dashboards.

6.2.1 Findings based on literature review

The literature review provided useful knowledge and understanding to the development of this project. The review found that inventory management is an important process in ensuring that business operations are running smoothly. Poor inventory management can result in many issues such as overstocking, dead stock, expired materials and stockouts. These issues may increase operational costs and impact customer satisfaction.

The review also discovered that many organizations have trouble trying to manage data sourced from different sources. This can result in slower and less accurate results. Previous studies suggested that a data engineering pipeline can enhance data quality and make data more accessible for analysis. Cloud technologies and business intelligence tools can also help organizations to process large amounts of data efficiently and present insights through interactive dashboards. The findings supported the use of Microsoft Azure and Power BI in the development of the proposed system.

6.2.2 Achievement of Project Objectives

The first objective of this project was to analyze the inventory risk management problem domain and review existing data engineering approaches for recoverable assets, dead stock and stockout risk detection. The objective was successfully reached through literature review, discussions with industry partners and analysis of the datasets provided. The study helped to identify the key business requirements and inventory risks to be addressed.

The second objective was to design a three-layer data pipeline architecture consisting of Bronze, Silver and Gold layers. A Medallion Architecture was designed to achieve the

objective of data ingestion, transformation, storage and analytics. The architecture was also designed with the business rules to identify RA and MC materials.

The third objective was to implement the proposed end-to-end data pipeline on Microsoft Azure. This was successfully achieved by using Azure Data Factory for data ingestion and transformation, Azure Data Lake Storage Gen2 for data storage and Azure Synapse Analytics for storage of analytics ready datasets. The pipeline was able to process data from different sources and generate meaningful outputs for analysis.

The fourth objective was to use Microsoft Power BI to create and evaluate an interactive dashboard. This objective was also met through the development of three dashboards: Executive Overview Dashboard, Inventory Analysis Dashboard and Sales Impact Dashboard. These dashboards provide insights on dead stock, expired materials, recoverable assets, inventory risks, stockout risks and affected sales orders.

Overall, the project was successful in developing an end-to-end data engineering solution for inventory risk management. By converting raw operational data into useful business insights, the system helps users monitor inventory conditions, identify potential risks and make better decisions.

6.3 System Constraint

The proposed data pipeline is designed to support dashboard visualisation for materials below reorder level, Magna Carta Dead Stock, Recoverable Assets, Magna Carta Expired Stocks and Delayed Sales Orders. Several constraints were identified during the design and implementation of the system.

6.3.1 Data Source Constraints

The system relies on six static CSV files provided by the stakeholder as the source of data. As no operational database or transactional system is connected to the pipeline, the accuracy and completeness of the outputs depend entirely on the quality of the files. Any missing, incorrect, or outdated records will propagate through the Bronze, Silver, and Gold layers.

Furthermore, the dataset represents a snapshot of business operations rather than a

continuously updated source. Consequently, the dashboards are limited to the information available in the provided files and cannot provide real-time insights.

6.3.2 Business Assumption Constraints

Several business assumptions were required due to incomplete information in the source data.

The quantity per unit in the Bill of Materials (BOM) is estimated by dividing total material consumption by total delivered sales quantity. Since actual production output quantities were not provided, the resulting BOM may differ from real production recipes.

In addition, it is assumed that all production orders are performed either to fulfill sales orders or replenish inventory, and that all available materials are reserved sequentially according to sales order priority. These assumptions directly influence the calculations for material shortages and delayed sales orders.

6.3.3 Transformation and Computation Constraints

The Gold layer contains multiple business calculations, including adjusted lead time, provision value, recoverable assets, running material consumption, material risk classification, and delayed sales order determination.

Implementing these calculations within Azure Data Factory introduces limitations because Data Factory is primarily designed for orchestration and ETL operations rather than advanced analytical computations. As a result, some complex business rules are difficult to implement and maintain, increasing the risk of inaccurate calculations within the Gold layer.

This limitation was observed during implementation, where several Gold-layer computations required extensive transformation logic and produced calculation inconsistencies.

6.3.4 Security Constraints

The data contains sensitive business information, including inventory levels, customer orders, supplier information, material costs, and Bill of Materials (BOM) data. Role-based access control is not currently implemented within the system. Consequently,

access management relies on the underlying Azure services and operational controls rather than data-level security.

6.3.5 Reporting Constraints

The dashboards are dependent on the accuracy of the derived Gold-layer datasets. Any errors in business rules, assumptions, or transformation logic will directly affect the reported values for Magna Carta Dead Stock, Recoverable Assets, and Delayed Sales Orders.

Since the dashboards are designed for analytical reporting purposes only, they are not intended to support operational decision-making that requires real-time inventory or order status updates.

6.4 Suggestions for Future Improvement

Although the proposed RA and IRM system has achieved its main objectives, several improvements can be considered to strengthen future implementation and execution. Since the current system still depends on provided CSV datasets and dashboard reporting, future work should focus on automation, real-time integration, stronger validation, and better user adoption.

6.4.1 Data Integration and Automation

Future implementation should reduce dependency on manually uploaded CSV files by connecting the system directly to live operational data sources such as ERP systems, inventory databases, purchase records, sales records, and production data. This will allow the pipeline to receive updated data more consistently. In addition, Azure Data Factory triggers can be used to schedule automatic pipeline execution, such as daily or weekly refresh. This helps reduce manual work and ensures that the Bronze, Silver, and Gold layers are always updated for reporting.

6.4.2 Data Quality and Business Rule Enhancement

Future improvement should include stronger data quality validation before the data is used for analysis. The system should check missing values, duplicate records, invalid dates, incorrect material codes, and abnormal stock quantities. Besides that, the business rules in the Gold layer can be improved further, especially for Recoverable Assets

detection, Magna Carta dead stock, expired inventory, stockout risk, and financial risk calculation. This will make the system more reliable and easier to maintain when business rules change.

6.4.3 Dashboard Improvement and Deployment Plan

The Power BI dashboard can be improved by adding more useful features such as detailed material drill-through, risk trend analysis, alert indicators, and clearer summary KPIs for management users. User feedback from inventory managers and supply chain managers should also be collected to ensure that the dashboard supports real decision-making. For implementation, the project can be deployed in stages: first validate the pipeline using historical data, then connect it to regularly updated data sources, improve the dashboard based on feedback, and finally deploy it for regular business use with proper access control and monitoring.

6.5 Summary

This chapter summarized the overall achievements of the project which assessed the level of success in achieving the project goals, highlights any major constraints faced whilst developing the project, and suggests directions of future improvements. The literature review revealed that the problems associated with poor inventory management include overstocking, dead stock, expired materials, stock outlines, resulting in a negative effect on customer satisfaction and the increase in operational expenses. The results validated the proposed solution by using Microsoft Azure and Power BI as the underlying technologies for the proposed solution. The four project objectives all met the requirements which includes inventory risk management problem domain analysis, design of a three layer Medallion Architecture, implementation of end-to-end data pipeline on Microsoft Azure, and creation of three interactive Power BI dashboards.

There were some constraints found throughout the project, including data that is from static CSV files only provided for real-time insights, business assumptions had to be made as of the limited availability of source data, computation difficulties in Azure Data Factory with complex Gold layer logic, and lack of role-based access control. There is a possibility to integrate the system with operational data sources with automated pipeline scheduling for future improvements, while providing more robust data quality validation, business rule and dashboard capabilities. It is also suggested that a phased deployment

strategy be considered with appropriate access control and feedback from the users integrated into the system to ensure that the system is effectively used for regular business operations.

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